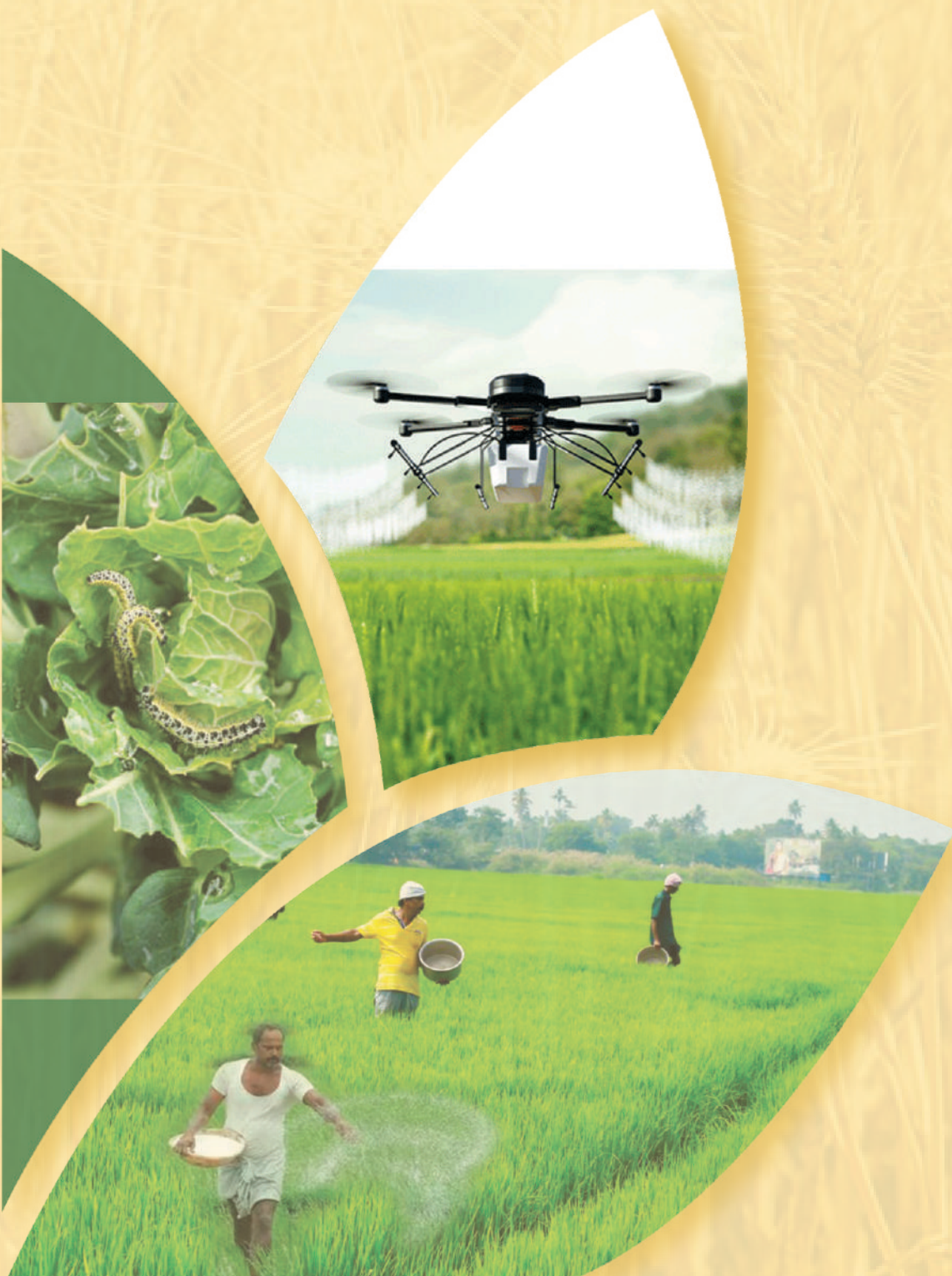


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Editorial

Advancing the Frontiers of Plant Health: A Call for Technological Synergy

It is with great pride and a sense of professional fulfillment that I present the March 2026 issue of the *Indian Journal of Plant Protection* (IJPP). This month marks a significant milestone for our community, following the highly successful National Conference organized by the Plant Protection Association of India (PPAI) in collaboration with the National Institute of Plant Health Management (NIPHM), during Feb 26 to 28, 2026.

The conference served as a vibrant platform for scientists, policymakers, and industry experts to deliberate on emerging threats to global food security. PPAI's pivotal role in this convention underscored our society's 54-year legacy as a cornerstone of plant health advocacy in India. The discussions highlighted a critical consensus: the future of plant protection lies in the seamless integration of traditional wisdom with disruptive modern technologies.

In this current issue, we feature seven full-length research articles and one short communication, representing the rigorous scholarship our members are known for. This edition is particularly strong in Plant Pathology, with five high-quality papers exploring disease dynamics, as well as two insightful contributions from the field of Entomology. Together, these works provide essential data for managing the evolving pest and disease landscapes.

However, as we witness a paradigm shift on the ground, the IJPP Editorial Team wishes to draw the attention of our members, scholars, and students toward the next frontier. We strongly encourage the submission of research focusing on:

- Artificial Intelligence (AI) and Machine Learning (ML) for high-precision pest forecasting and diagnostics.
- CRISPR-enabled technologies and RNAi-related research for targeted pest management.
- Advanced studies in Biocontrol and sustainable ecological interventions.

The IJPP is committed to evolving alongside these technologies. Our editorial team is working tirelessly to reduce the “submission-to-publication” turnaround time. To assist us in this goal, I earnestly request all authors to strictly adhere to the IJPP guidelines and formatting protocols. Compliance at the submission stage is the single most effective way to ensure your valuable research reaches the global scientific community swiftly.

As we look ahead, I invite our members and well-wishers to continue submitting their highest-quality research. Our 54-year-old society thrives on the intellectual contributions of its stakeholders, ensuring we remain a global leader in plant protection sciences.

I extend my deepest gratitude to the Executive Committee of the PPAI and our dedicated members for their unwavering support and cooperation in bringing this issue to fruition.

Prof T V K Singh

Chief Editor

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Kusmi lac production in Central Narmada Valley, Madhya Pradesh, facts and factors

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Abstract

The study was conducted in district Narmadapuram of the state of Madhya Pradesh, India during the year 2017-2018. This district is located in the 'Central Narmada valley' agro-climatic zone and is known for its agricultural production. In order to gather data, seven villages were selected from the district, with ten farmers from each village chosen for the study. After analyzing the data, it was found that the fresh and dry weight of the Jethwi crop, Oleosa tree was significantly higher than that of Agahni crop. However, despite this difference, it was observed that this attribute did not contribute to an increase in overall yield from the tree. Interestingly, it was noted that the yield per plant and yield per foot stick were higher in Jethwi crop, but these differences were not statistically significant.

Keywords: Lac, Aghani, Jethwi, Kusmi, Kerria lacca, Tachardiidae, Homoptera

Introduction

Lac is a natural resin that is produced by a tiny insect called *Kerria lacca* (Kerr), belonging to the family Tachardiidae (Homoptera) (Meena *et al.*, 2022). This insect feeds on plant sap and secretes a resinous substance (lac) on the branches of its host tree (Singh *et al.*, 2009), which hardens when it comes in contact with air. The adult lac insects are protected by a hard lac crust, and the resin they produce is completely natural, biodegradable, and non-toxic. As a result, it is widely used in various industries including food, textiles, and pharmaceuticals (Chen *et al.*, 2010). The lac insect has been found to have around 113 host tree species and can complete its life cycle on them (Kumar, 2013). However, only three traditional host plants - Kusum (*Schleichera oleosa*), Palas (*Butea monosperma*) and Ber (*Ziziphus mauritiana*) are used for commercial lac cultivation. These host plants are specifically chosen for their ability to provide the necessary nutrients for the lac insect to thrive and produce high-quality resin. In India, there are two major strains of the lac insect known

as *Kusmi* and *Rangeeni*, each with specific host plants. *S. oleosa* is the suitable host for the *Kusmi* strain while *B. monosperma* is suitable for the *Rangeeni* strain. However, *Z. mauritiana* is a common host for both strains of the lac insect (Kumar, 2013).

The *Kusmi* lac insect has played a crucial role in the production of lac crops in India, accounting for 25% of the total production (Anees and Monobrullah 2016). This particular strain of lac insect is highly prized for its unique ability to produce two crops in a year - the summer crop or *Jethwi* crop and the winter season crop or *Aghani* crop. These crops require six months each to mature, making it important to carefully consider the suitable host plant and climatic conditions when starting lac cultivation.

As the state of Madhya Pradesh is situated in the central part of India, it holds a significant position in the production of *Kusmi* lac. During the year 2013-14, Madhya Pradesh contributed to 2,497 tons of the total 21,008 tons of lac

produced in India. The production of *Kusmi* lac, Madhya Pradesh is mainly dependent on *S. oleosa* trees, which are abundantly available in the state. With the availability of these host trees and favorable climatic conditions, the state has been able to achieve high production levels of *Kusmi* lac. In order to further understand the production capabilities of *Kusmi* lac, the present study was conducted in district Narmadapuram, Madhya Pradesh on *S. oleosa* host tree. This study also aimed to understand the seasonal variations and production capabilities of *Kusmi* lac in Madhya Pradesh.

Materials and methods

The study was conducted in district Narmadapuram of the state of Madhya Pradesh, India during the year 2019-20. The district Narmadapuram comes under 'Central Narmada valley' agro-climatic zone. Seven villages were selected from the district and ten farmers were chosen from each village for data collection. The study recorded data such as lac yield per tree, lac yield per foot lac stick, and dry weight per 100 cells from each farmer. The data was subjected to statistical analysis through the t-test.

Table 1. Agro-climatic zone, district and villages

Agro-climatic zone	Name of district	Name and no. of village covered	No. of Lac growers covered
Central Narmada valley	Narmadapuram	(1) Kamti, (2) KhapaKalan, (3) Vijnanmhigh, (4) Dangahigh, (5) Dhadav, (6) Padav and (7) Murgidhana	100

Results and discussion

Data collected from seven villages of district Narmadapuram was analyzed and described under following head.

Lac production per plant

There was no considerable variation in lac production per plant between the *Aghani* and *Jethwi* cropping seasons. The recorded production in the *Aghani* season was 36.033 kg/plant, while in the *Jethwi* season it was slightly higher at 40.067 kg/plant. This indicates that the cultivation practices used during both seasons were similar and had a like impact on lac production. Therefore, it can be inferred that if the cultivation practices remain consistent, the lac production would be at par in both seasons.

Table 2. Avg. Lac production/plant (In kg.)

S. No.	Village	<i>Aghani</i>	<i>Jethwi</i>
1	Kamti	36.6	38.7
2	Khapa Kalan	40.6	42.7
3	Vijnanmhigh	38.8	40.7
4	Dangahigh	41.2	39.3
5	Dhadav	32.5	39.0
6	Padav	26.5	40.0
7	Murgidhana	25.8	28.6
Mean		36.033	40.067

Table 3. t-Test: Two-Sample Assuming Equal Variances

Session	<i>Aghani</i>	<i>Jethwi</i>
Mean	34.571	38.429
Variance	41.442	20.606
Observations	7.000	7.000
Pooled Variance	31.024	
Hypothesized Mean Difference	0.000	
df	12.000	
t Stat	-1.296	NS
P(T<=t) one-tail	0.110	
t Critical one-tail	1.782	
P(T<=t) two-tail	0.220	
t Critical two-tail	2.179	

Lac yield per foot stick lac

Jethwi crop yielded higher per foot stick that was 50.28 g, than that of *Aghani* season (44.2 g/ higher lac per foot stick), however, the yield difference was not significant. This suggests that factors other than the crop season may also play a role in determining the yield of lac. Further research is needed to identify these factors and understand their impact on lac yield. Nonetheless, the data highlights the potential of *Jethwi* crop for producing high yields and could be further explored for commercial cultivation.

Table 4. Lac yield per foot stick lac (In g.)

S. No.	Village	Aghani	Jethwi
1	Kamti	40.56	53.08
2	Khapa Kalan	38.72	42.31
3	Vijanmhigh	34.96	45.65
4	Dangahigh	48.82	55.53
5	Dhadav	51.62	55.58
6	Padav	48.64	49.10
7	Murgidhana	46.33	50.28
	Mean	44.2	50.2

Table 5. t-Test: Two-Sample Assuming Equal Variances

Session	Aghani	Jethwi
Mean	44.236	50.219
Variance	38.217	24.969
Observations	7.000	7.000
Pooled Variance	31.593	
Hypothesized Mean Difference	0.000	
df	12.000	
t Stat	-1.991	NS
P (T<=t) one-tail	0.035	
t Critical one-tail	1.782	
P (T<=t) two-tail	0.070	
t Critical two-tail	2.179	

Fresh and dry weight of 100 lac cells.

The analysis of fresh and dry weight of *Jethwi* and *Aghani* crops revealed that there was a significant difference in the fresh weight of the two crops, with *Jethwi* produce having a higher fresh weight (7.09 gm/100 cell) compared to that of *Aghani* (6.82 gm/100 cell). The trend was similar in the dry weight measurements as well, with *Jethwi* crop showing a higher value of 6 gm/100 cell compared to *Aghani* crop's 5.69 gm/100 cell. These findings indicate that *Jethwi* crop has a higher overall biomass production compared to *Aghani* crop, which could be attributed to various factors such as genetics, soil quality, and environmental conditions. (Mohanta *et al.*, 2014) also reported the higher cell weight in *Jethwi* crop, 17.0 to 21.40 mg/cell in winter season and 19.00- 25.60 mg/cell in summer season.

Table 6. Mean fresh and dry weight of 100 lac cells (In gm)

S. No.	Village	Aghani		Jethwi	
		Fresh weight	Dry weight	Fresh weight	Dry weight
1	Kamti	6.59	5.25	6.5	5.43
2	Khapa Kalan	7.21	6.26	7.23	6.13
3	Vijanmhigh	6.65	5.45	7.51	6.52
4	Dangahigh	6.97	6.1	7.08	6.03
5	Dhadav	6.47	5.35	7.45	6.67
6	Padav	6.48	5.76	6.25	5.38
7	Murgidhana	7.4	5.68	7.64	5.84
	Mean	6.82	5.69	7.09	6.00

Table 7. t-Test: Two-Sample Assuming Equal Variances

100 cell weight		Fresh weight		Dry weight	
Session		Aghani	Jethwi	Aghani	Jethwi
Mean		6.82	5.69	7.09	6.00
Variance		0.14	0.14	0.28	0.25
Observations		7.00	7.00	7.00	7.00
Pooled Variance		0.14		0.26	
Hypothesized Difference	Mean	0.00		0.00	
Df		12.00		12.00	
t Stat		5.63	**	3.99	**
P (T<=t) one-tail		0.00		0.00	
t Critical one-tail		1.78		1.78	
P (T<=t) two-tail		0.00		0.00	
t Critical two-tail		2.18		2.18	

Conclusion

The *Kusmi* lac strain of *S. oleosa* was found to perform equally well in both *Aghani* and *Jethwi* seasons, with no significant differences in overall yield. However, when comparing the fresh and dry weight of the crop, it was discovered that the *Jethwi* season resulted in significantly higher weights. Despite this difference, it can be concluded that the *Kusmi* lac strain demonstrates consistent performance in both seasons. These results also highlight the adaptability and resilience of the *Kusmi* lac strain, making it a valuable resource for lac production in various environments.

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Management of *Alternaria* leaf disease of groundnut using fungicides

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Abstract

A field experiment was carried out to evaluate the efficacy of different fungicides in managing *Alternaria* leaf spot and blight in peanut. The results indicated that all fungicidal treatments significantly reduced disease severity compared to the untreated control, with no statistical differences among treatments. Among the tested fungicides, two sprays of Difenoconazole @ 0.025% at 60 and 70 days after sowing (T5) were most effective, recording the lowest disease severity (3.3) against 4.8 in control with the highest per cent efficacy of 32%. Biological, pod, and haulm yields were significantly enhanced by all treatments, with T5 showing the maximum pod yield (1807 kg/ha) and haulm yield (3162 kg/ha) compared to control (1572 and 2752 kg/ha, respectively), resulting in a 13% increase in biological yield. Economic analysis revealed that T5 treatment achieved an incremental cost-benefit ratio (ICBR) of 1:4.8, slightly lower than T1 (1:6.1) due to higher fungicide cost. Overall, Difenoconazole @ 0.025% proved to be the most effective treatment for the management of *Alternaria* leaf spot and blight in peanut with significant yield improvement.

Keywords: *Alternaria* leaf blight, *Alternaria* leaf spot, Peanut, Fungicides, disease management

Introduction

Groundnut crop is prone to attack by diseases to a much larger extent than many other crops. Among the foliar fungal diseases, *alternaria* leaf spot (*Alternaria alternata* Keissler), and *alternaria* blight (*Alternaria tenuissima*) are of economically important. Though, the leaf spot disease caused by *A. alternata* (Fr.) Keissler, was reported to occur two decades ago by Balasubramanian (1979), Subrahmanyam *et al.* (1981), Vyas *et al.* (1985) and Narain *et al.* (1987). Patil and Hiremath (1989) and Ghewande *et al.* (1982) reported *A. tenuissima* (Kunze. Fr) Wiltshire causing leaf blight disease, while two other species of *Alternaria* reported from groundnut were described by Kulkarni (1974) with brown leaf spot

symptoms and observed to cause appreciable damage to the crop, as well as *A. longipes* described by Giri and Murugesan (1996) causing necrotic leaf spots. Aulakh (1969) reported a leaf blight of groundnut caused by *A. tenuis* Auct. The disease was found very destructive (40 to 50% damage to the foliage) in the Punjab. The leaf blight disease caused by different species of *Alternaria* has been hitherto a minor disease in Gujarat until an outbreak of *Alternaria* leaf blight was observed in summer crop of groundnut in farmer's fields of Saurashtra and Kutch regions of Gujarat during 2009. Kumar *et al.* (2012). The losses due to *Alternaria* leaf blight have been reported varying from 13 to 22% and

the reduction in haulm yield from 24 to 63% depending upon the variety.

Fungicides are the last line of defense in the armory of an integrated disease management (IDM) approach. They do not create yield, but protect an inherent yield potential that the grower may realize in the absence of disease. In the field, securing effective disease control from fungicide applications is dependent upon the disease pressure and the effectiveness of the fungicide to control that disease (Poole and Arnaudin, 2014). Woodward *et al.* (2013) reported that pesticides have been used in some form for centuries, with sulfur compounds and botanicals being used as early as 2500-1500 BC by the Sumerians and Chinese for insect and disease control. Sulfur, copper, and organic mercury were used in the 1800s as fungicides, and Bordeaux mix has been used since 1885 for powdery and downy mildew control. Since that time and beginning in the late 1930s to the present day, tremendous activity has occurred related to the development of synthetic and biological based pesticides numerous chemical fungicides are available for control of the groundnut-diseases. Applications of these products are made for the management of both foliar and soil-borne diseases. These products have traditionally been the second largest variable expense in groundnut production, behind seed cost.

Material and methods

An experiment was conducted at the research farm of Directorate of Groundnut Research, Junagadh, Gujarat (70° 36' E longitude and 21° 31' N latitude and at an altitude of 60 m above mean sea level). The climate of the region is semi-arid with a mean annual rainfall of 850 mm. The rainfall is mainly confined during June to September months. The monthly average temperature is minimum in January (12 °C) and maximum in May (42°C). The soils under study were shallow (<30 cm), very dark grey in colour having clay texture (>40%), medium black and calcareous in nature underlain by weathered or hard milliolithic limestone parent materials, having good drainage facility. The surface soils (0-15 cm) were

alkaline (pH 8.2) in reaction, medium in organic carbon (0.49%) and phosphorus (10 kg ha⁻¹), low in nitrogen (130 kg ha⁻¹), high in potassium (167 kg ha⁻¹) and micronutrients were in the range of medium to high. The experiment was conducted on a fixed site for two years (twice) from 2010 to 2011 in the *rabi*-summer season, using randomized block design with seven treatments (Table 1) including appropriate control with four replication. The plot size was maintained at 25 m² (5 m x 5 m) plot size with 45 cm x 10 cm spacing and the groundnut variety (JAL - 42) was selected for the study.

Observations were recorded on severity of alternaria leaf diseases after 45 and 70 DAS (Days after sowing) by adopting a 1-9 rating scale as proposed by Kumar *et al.* (2012). Per cent efficacy of disease control (PEDC) and per cent yield increase over disease control (PIDC) worked out by using the following formulae:

Per cent efficacy over Disease Control (PEDC) = $\frac{[(\text{Disease severity or incidence in control} - \text{Disease severity or incidence in treatment}) / \text{Disease severity or incidence in control}] \times 100}{100}$

Per cent yield increase over Disease Control (PIDC) = $\frac{[(\text{Yield in treatments} - \text{Yield in Control}) / \text{Yield in Control}] \times 100}{100}$

The incremental cost benefit ratio (ICBR) and Benefit:Cost Ratio (B:C ratio) of all the treatments was calculated using following formulae:

$$\text{ICBR} = \frac{\text{Additional income received (from the particular treatment)}}{\text{Additional cost incurred (for the particular treatment)}}$$

$$\text{B:C ratio} = \frac{\text{Income received (from the particular treatment)}}{\text{Cost incurred (for the particular treatment)}}$$

The data were analysed following standard statistical procedures using DSAASAT software (Onobri, 2007) and analysis of variance (ANOVA) was worked out. Data was subjected to F test and critical difference (CD) between means was calculated at 5% level of probability ($p < 0.05$).

Table 1. Treatment applied for the management of *Alternaria* leaf diseases

Treat	Treatment Details
T ₁	Two spray of Mancozeb @ 0.2% at 60 & 70 DAS
T ₂	Two spray of Copper oxychloride @ 2% at 60 & 70 DAS
T ₃	Two spray of Hexaconazole @ 0.0075% at 60 & 70 DAS
T ₄	Two spray of Tebuconazole @ 0.375% at 60 & 70 DAS
T ₅	Two spray of Difenconazole @ 0.025% at 60 & 70 DAS
T ₆	Two spray of Propiconazole @ 0.025% at 60 & 70 DAS
T ₇	Control

Result and discussion

The data shown in Table-2 revealed that all the treatments/fungicides applied for the management of *Alternaria* leaf spot and blight significantly reduced the diseases in comparison to untreated control while all the treatments were at par. The treatment T5 (two spray of Difenconazole

@ 0.025% at 60 and 70 DAS) was the most effective among all the fungicides applied for reducing the ALS and severity of ALS was recorded 3.3 in comparison 4.8 in to untreated control. The per cent efficacy was also highest T5 (32%). Similar results were reported by Sunkad *et al.* (2005) where bio-efficacy of six fungicides in the control of leaf spots and rust of groundnut (JL-24) and their effect on dry pod and fodder yield was studied during *kharif*, 2001 and 2002. In the present study, the maximum disease control with higher dry pod and fodder yield was observed in hexaconazole, difenoconazole and propiconazole treatments. Effectiveness of chlorothalonil treatment was also better in comparison to mancozeb and carbendazim, while carbendazim was found fairly effective against leaf spots, and mancozeb showed better control of rust. Bhaskar *et al.* (2012), also reported that for controlling the Tikka disease pathogens, Difenconazole + Monocrotophos (0.025% + 0.05%) treatment showed minimum per cent disease intensity (PDI) of 30.97%.

Table 2. Disease severity of *Alternaria* leaf spot/ blight on JL-42 groundnut variety following treatment by fungicides and their efficacy

Treatment	Initial plant population			Alterneria leaf spot (using 1-9 scale)			Per cent Efficacy over Disease Control					
	2010	2011	Pooled	2010	2011	Pooled	2010	2011	Pooled			
T1	401	518	460	3.2	b	3.5	b	3.3	b	38.7	22.2	31
T2	411	546	479	3.4	b	3.3	b	3.3	b	33.8	27.8	31
T3	433	489	461	3.3	b	3.3	b	3.3	b	37.1	25.9	32
T4	406	542	474	3.4	b	3.3	b	3.4	b	33.8	25.9	30
T5	407	516	462	3.5	b	3.5	b	3.5	b	32.4	22.2	28
T6	400	551	476	3.8	b	3.5	b	3.6	b	27.4	22.2	25
T7	393	480	436	5.2	a	4.5	a	4.8	a	0.0	0.0	0
S.Em.±	10.01	11.70	16.37	0.40		0.22		0.23				
CD at 5%	29.75	34.78	46.96	1.18		0.66		0.65				

*Pooled mean of two experiments with four replication for the year 2010 and 2011

**Means followed by the same letter in a column are not significantly different at $p = 0.05$ according to Duncan's multiple range test

The biological yield was significantly increased in all the treatments over control (Table 3). The highest pod yield and haulm yield were observed in T5 (1807 kg/ha and 3162 kg/ha, respectively) in comparison to control (1572 kg/ha and 2752 kg/ha, respectively). There was an increase of 13% biological yield in T5 over control. Similar results were reported by Bhaskar *et al.* (2012) where combination of Difenconazole + Monocrotophos (0.025% + 0.05%) showed maximum pod (2250 kg/ha) as well as fodder yields (6451 kg/ha).

Incremental cost benefit ratio was 1:4.8 in case of T5 lesser than that of (1:6.1) in T₁, due to higher price of the fungicide (Table 4). Sunkad *et al.* (2005) also observed high incremental cost benefit ratio (JCBR) with hexaconazole (1:4.72) followed by propiconazole (1:2.05), difenoconazole (1:1.29) and chlorothalonil (1:1.13). Similarly, the data on the incidence of foliar diseases of groundnut recorded by Haque *et al.* (2008), Hagan *et al.* (2010), Grichar *et al.* (2010 a, and b). Augusto and Brenneman (2011), Rahman *et al.* (2014).

Table 3. Pod and Haulm Yield [kg/ hectare (ha)] of groundnut treated with fungicide for ALS/ALB control

Treatment	Pod Yield kg/ha						Haulm Yield kg/ha						Per cent Increase over Disease Control		
	2010		2011		Pooled		2010		2011		Pooled		2010	2011	Pooled
T1	1375	a	1972	bc	1673	bc	2406	a	3451	bc	2929	bc	3.0	8.1	6.0
T2	1350	a	2024	b	1687	b	2363	a	3543	b	2953	b	1.3	10.5	6.8
T3	1358	a	1958	bc	1658	bc	2376	a	3426	bc	2901	bc	1.8	7.5	5.1
T4	1376	a	2238	a	1807	a	2408	a	3916	a	3162	a	3.1	19.0	13.0
T5	1375	a	2090	ab	1733	ab	2406	a	3658	ab	3032	ab	3.0	13.3	9.2
T6	1375	a	1904	bc	1640	bc	2406	a	3333	bc	2869	bc	3.0	4.9	4.1
T7	1333	a	1812	c	1572	c	2333	a	3170	c	2752	c	0.0	0.0	0.0
S.Em.±	43.31		58.50		63.83		75.80		102.38		111.70				
C.D. at 5%	128.69		173.83		183.07		225.21		304.19		320.65				

*Pooled mean of two experiments with four replication for the year 2010 and 2011

**Means followed by the same letter in a column are not significantly different at $p = 0.05$ according to Duncan's multiple range test

Conclusion

In the present study all the fungicidal treatments effectively reduced *Alternaria* leaf spot and blight severity compared to the untreated control. Among them, two sprays of difenoconazole at 0.025% applied at 60 and 70 DAS proved most effective in minimizing disease severity and enhancing both pod and haulm yields. Although the incremental cost-benefit ratio of T5 was slightly lower due to higher chemical cost, its effectiveness for superior disease control and yield improvement justify its use. Therefore, difenoconazole at 0.025% may be recommended as an efficient and economically viable option for the management of *Alternaria* leaf diseases in groundnut cultivation.

Table 4. Economic feasibility of fungicides applied for ALS/ALB control

Treatment	B:C ratio			ICBR		
	2010	2011	Pooled	2010	2011	Pooled
T1	1.8	2.4	2.1	1:3.3	1:8.8	1:6.1
T2	1.7	2.4	2.1	1:1.8	1:8.3	1:5.1
T3	1.7	2.4	2.1	1:2.0	1:6.7	1:4.3
T4	1.7	2.6	2.2	1:1.8	1:7.8	1:4.8
T5	1.7	2.4	2.0	1:1.0	1:3.8	1:2.4
T6	1.7	2.3	2.0	1:1.7	1:3.1	1:2.4
T7	1.7	2.2	2.0	-	-	-

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Assessing the impact of *Pseudocercospora* leaf spot infection on commercially important silkworm cocoon characters

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Abstract

One of the major challenges to successful sericulture in India is the prevalence of infections in mulberry. In West Bengal, *Pseudocercospora* leaf spot, which is caused by [*Pseudocercospora mori*], is a highly frequent ailment. An investigation was conducted to determine the impact of feeding mulberry leaves infected with *Pseudocercospora* leaf spot on the number and quality of cocoon and silk. The silkworm larvae were fed varying grades of *Pseudocercospora* leaf spot infected leaves, and the rearing parameters (larval duration, length, weight, single cocoon weight, single shell weight, shell ratio, ERR wt., ERR no., mortality and biochemical analysis of leaves) were used to measure the effects of the feeding. The total larval duration increased by 4 more days due to the feeding of different graded of *Pseudocercospora* leaf spot leaves in comparison to 22 days in control. Decreased from 3.13 g to 0.65 g (G-IV). The larvae weight from 2.00g to 0.85g for single cocoon weight; from 0.44 g to 0.09 g for single shell weight; from 22.02% and 11.04% for shell ratio percentage; from 19.53 kg to 1.58 kg for effective rate of rearing weight and from 9266 to 1400 for effective rate of rearing number in respect of healthy and Grade-IV diseased leaves. The reduction in quality and yield of larvae was proportional to the level of disease severity. It was observed that all the biochemical properties i.e. amount of moisture, protein, chlorophyll and total sugar contents were reduced drastically in mulberry leaves due to *Pseudocercospora* leaf spot diseases which negatively affect the growth of larvae and cocoon.

Keywords: *Bombyx mori*, *Pseudocercospora* leaf spot, *Morus* spp., economic parameters, biochemical analysis.

Introduction

Sericulture is considered a significant labour-intensive, low-capital-investment agro-based cottage industry that has the potential to employ more than six million people in India. The quality and quantity of mulberry leaves used as food are known to affect the growth and development of silkworms (*Bombyx mori* L.), which is in turn determined by the economic properties of the cocoon crop. The several biotic diseases that affect mulberry, *Myrothecium* leaf spot, *Pseudocercospora* leaf spot, Bacterial leaf spot, and Powdery mildew are highly prevalent and cause farmers in West Bengal to suffer significant huge losses. Mulberry pathologists might therefore believe that it's crucial to

describe crop loss in terms of cocoon crop loss, poor cocoons generated at one point in time, or overall financial loss on the other hand. Mulberry leaf protein plays the key role in larval growth, silk gland development, cocoon production and cocoon quality (Bongale and Chaluvachari 1995), which has been reflected from several studies conducted on varietal performance (Mishra *et al.*, 1996; Ram Rao *et al.*, 2000). The protein content increases with age of leaf (Jha *et al.*, 2016) but contradictory claim also exists (Matei *et al.*, 2006). Dandin *et al.* (2000), reported the feeding of diseased leaves adversely affects the growth and health of silkworm resulting in partial or sometimes

complete failure of cocoon crops and reduced the marketing quality of produced cocoons. Parameters like larval weight, mortality per cent, single cocoon weight, single shell weight and other economic parameters were considered (Gangwar 2010, Jha *et al.*, 2016) and calculation should be made at three different seasons separately. According to Qadri *et al.* (1999), poor quality cocoon and the maximum disease severity of 26% resulted in an estimated 54-58 per cent crop loss. According to Tang *et al.* (2005), the only source of nutrients for silkworm growth is the mulberry leaves, which contain protein, carbs, vitamins, minerals, and other elements. In order to determine the impact of diseased foliage on cocoon generation and other economic metrics, researchers typically only expose infected foliage to it for one or two instars. The actions of the larvae and their moulting during all earlier stages, such as Instar I, Instar II, etc., are what determine the development of cocoons. Ito (1978) emphasized the significance of diet in the Japanese silk industry. A cultivar with a lower sugar content and a higher protein content should have a longer larval life and a lower moulting ratio (Chaluvachari and Bongale 1996). Bakshi *et al.*, (1972) and Dutta *et al.* (2011) reported that infected mulberry leaves are not suitable for silkworm feeding causing poor cocoon. Manimegalai and Chandramohan (2007) noted that the primary cause of the higher yield of high-quality cocoons is the quality of the leaves. When damaged mulberry leaves were fed to silk worm larvae, there is a decrease in their fecundity and the proportion of eggs that hatch in the second generation. Except for ERR by number, all other rearing and reeling parameters have a substantial negative connection with the severity of the disease (Quadri *et al.*, 1999). A wide range of silk worm rearing parameters (larval duration, larval length, larval weight, single cocoon weight, single shell weight, shell ratio, ERR weight, ERR number, and mortality) were used to measure the effects of feeding larvae with different grades of diseased mulberry leaves start on the quality and quantity of cocoon as well as silk quality. In the study of feeding *Pseudocercospora* infected mulberry leaves fed to silkworms affects their metabolism, there is a chance that the production of cocoons might not succeed entirely or partially, or that it will result in poor cocoons with low market value. It's also true that lower-quality cocoons cause reeling quality to decline.

Material and methods

The experiment was conducted in the year 2019 at village, Poradanga (Latitude: N 24° 8' 33" and Longitude: E 87° 57' 27") Murshidabad district of West Bengal, India. Leaf samples were randomly selected from ten locations within a plot from each zone. In the S-1 mulberry variety, tender, medium, and coarse leaves were allowed to feed for distinct instars of silkworm rearing (Chawki and Late age stage). In each plot five plants in a plot were examined for disease severity of *Pseudocercospora* leaf spot; four of the plants were selected in the plot's four corners, and one plant was located in its center. Three branches were randomly chosen from each plant and tagged. On the chosen branches, the total number of leaves and the number of diseased leaves were counted in order to determine the disease incidence (PI). The following 0-5 grading scale was used to classify all infected leaves into different infection grades in order to calculate the percentage of disease index (PDI).

The cross breed of silk worm N X F1 was used in this study. From Grade I to V, the *Pseudocercospora* infected leaves were fed to the second instar silkworm larvae during the rearing phase. Each treatment was given three times in duplicate to a total of fifty larvae. Observations were taken on length, weight and duration of the larvae till they reached maturity. Also following the recommended protocol by Krishnaswami (1970), leaf samples were taken from mulberry plants that were naturally infected with Grades I and IV, as well as from healthy mulberry plants that were in the same field and at the same age. The signs and symptoms of infection on the leaf lamina were used to categorize diseased leaves (1-5 grade). Biochemical examination of these leaves was done through evaluation of parameters such as leaf moisture by oven drying method, total chlorophyll (Arnon, 1949). The rearing data was subjected to statistically by one way ANOVA (Completely Randomized design) with DMRT to determine significant differences among the treatments.

Result and discussion

The experiment revealed that the quality of mulberry leaves has a predominating influence on the development of silk worm larvae and infection due to *Pseudocercospora* leaf spot adversely affected the quality of cocoons. That the total

larval duration was increased by 4 days (26 days) due to the feeding of different graded *Pseudocercospora* leaf spot leaves to (22 days) in control. The maximum average larval length (7.27 cm) was noticed in control, while smaller size (5.38 cm average) was recorded in the larvae fed with grade-IV diseased leaves. The larvae fed with Grade-V leaves did not survived up to 5th instar. Responses were similar with other parameters and declined trends were noticed from healthy to higher graded diseased leaves with minimum

values in Grade-IV diseased leaves. The individual larval weight decreased from 3.13 g to 1.25 g (G-IV) from 2.00 g to 0.85 g for single cocoon weight; from 0.44 g to 0.09 g in case of single shell weight and from 22.02% to 11.04% for shell ratio percentage. The effective rate of rearing weight decreased from 19.53 kg to 1.58 kg and the effective rate of bearing number got reduced from 9266 to 1400 in the grading order from healthy to grade iv disease leaves (Table 1 and fig. 1).

Table 1: Effects of the feeding of *Pseudocercospora* leaf spot infected mulberry leaves on the commercially important cocoon characters of the silkworm.

Treatment	Healthy	G-I	G-II	G-III	G-IV	G-V	CD 1%	CD 5%
Avg. Larval duration (day)	22	23	24	25	26	26	-	-
Avg. Larval length(cm)	7.27	6.59	6.13	5.85	5.38	0.00	0.13	0.09
Avg. Larval wt. (g)	3.13	2.44	2.05	1.84	1.25	0.00	0.31	0.22
Avg. Single cocoon wt. (g)	2.00	1.71	1.21	0.94	0.85	0.00	0.17	0.12
Avg. Single shell wt. (gm.)	0.44	0.26	0.19	0.12	0.09	0.00	0.04	0.03
Avg. Shell ratio %	22.02	15.38	15.21	12.40	11.04	0.00	2.02	1.42
Avg. ERR wt. (kg)	19.53	12.65	7.42	4.09	1.58	0.00	1.44	1.01
Avg. ERR (No)	9266	6866	5266	3533	1400	0.00	843.3	592.9
Avg. Mortality %	4.66	26.67	39.33	59.33	82.0	100.	3.89	2.739

Effects of *Pseudocercospora* leaf spot mulberry diseases on the biochemical attributes of mulberry leaves:

The total moisture content in mulberry leaves was found to be influenced greatly with the infection of pathogens and the percentage of reduction increased with the higher grades of diseased leaves (Table 2). The reduction in moisture content 13.99% in Grade -I and 22.77% in Grade-IV diseased leaves. Chlorophyll content of diseased leaves also varied with the infection of different pathogens and percentage

reduction in chlorophyll content was 11.87% in Grade-I diseased leaves and 29.45% was observed in Grade -IV. Statistical analysis indicated no significant differences between the chlorophyll content of leaves of diseases. A range of 11.32 and 39.00% reduction in total protein was observed in Grade -I and Grade-IV, whereas the reduction in total sugar 17.32% Grade-I and 36.40% Grade-IV were observed in diseased leaves (Table 2).

Table 2: Effects on the biochemical attributes of mulberry leaves due to *Pseudocercospora* leaf spot disease.

Content	Healthy Leaf	Grade-I		Grade-IV	
		Total	Reduction (%)	Total	Reduction (%)
Total Moisture	75.26	65.48	13.00	58.12	22.77
Total Chlorophyll	4.55	4.01	11.87	3.21	29.45
Total Protein	20.58	18.25	11.32	12.45	39.00
Total Sugar	45.36	37.50	17.32	28.85	36.40

The experiment confirmed that reduction in quality and yield during rearing of silkworm was proportionate to the level of disease severity in the leaves used for feeding. Many silk worm rearing parameters (larval duration, larval length, larval weight, single cocoon weight, single shell weight, shell ratio, ERR weight, ERR number, and mortality) were taken into consideration in the study. It is a well-known fact that the cost of producing mulberry in India

is much higher (60%) than the cost of rearing silkworms for the production of cocoons (40%). Chaudhary and Giridhar (1987) pointed out the need for both qualitative (thickness, wax content, nutritional parameters, etc.) and quantitative (leaf yield, shoot biomass, etc.) improvements of mulberry among researchers and farmers. A number of variables, such as varietal behaviour (Arug, 1942; Ravikumar, 1988), soil status (Das *et al.*, 1993; Mishra and Sharma, 2004),

disease intensity (Das *et al.*, 2001; Govindaiah and Naik, 2005), etc., influence the nutritional quality of foliage. According to Monir *et al.* (2017), one of the main obstacles to successful sericulture in India is the incidence of several diseases in mulberries, of which *Myrothecium roridum* is particularly frequent and causes farmers to face large losses. Timely treatment of diseases is a crucial part of acquiring nutritious and fresh leaves, as feeding diseased leaves results in a poor cocoon crop and reduces the income of the rearers (Govindaiah *et al.*, 2005).

Conclusion

Feeding of *Pseudocercospora* leaf spot leaves have detrimental effects on the silk larvae, cocoon quality as well as on silk. Most of the larvae have not been able to survive after the feeding of Grade -V diseased leaves and the health and quality of cocoon deteriorated due to the feeding of different grade of diseased leaves. Therefore, feeding of quality leaf is of utmost importance to get proper quality of cocoon and silk. Further it was observed that all the biochemical properties *i.e.* amount of moisture, protein, chlorophyll and total sugar contents were reduced drastically in mulberry leaves due to diseases which negatively affect the growth of larvae and cocoon.

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Evaluation of triazole and strobilurin group fungicides and their combinations against rice leaf blast under nursery conditions

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Abstract

Rice blast, caused by Magnaporthe oryzae, is a major constraint to rice production. A field experiment was conducted under nursery conditions to evaluate the efficacy of triazole, strobilurin, and their combination fungicides along with a biocontrol agent (Pseudomonas fluorescens) for the management of rice leaf blast. Significant differences were observed among treatments in reducing disease severity. The lowest Per cent Disease Index (PDI) was recorded with Trifloxystrobin + Tebuconazole @ 0.08% (13.33%), which resulted in the highest disease reduction (77.20%) over control. This was followed by Tricyclazole @ 0.06% (14.41%) and Pyraclostrobin @ 0.04% (18.14%), which were statistically on par. Combination fungicides such as azoxystrobin + tebuconazole and Azoxystrobin + Difenconazole showed moderate efficacy. Individual triazoles, Tebuconazole and Difenconazole, were comparatively less effective. The biocontrol agent Pseudomonas fluorescens recorded moderate disease suppression (PDI 35.90%) and was superior to control. The study highlights the effectiveness of combination fungicides, particularly Trifloxystrobin + Tebuconazole, in managing rice blast under nursery conditions.

Keywords: Rice leaf blast, Nursery, Fungicides, Triazoles, Strobilurins

Introduction

Rice blast caused by *Pyricularia oryzae* Cavara (The anamorph of *Magnaporthe oryzae* B.C. Couch), is one of the most destructive and widespread disease (Jia *et al.*, 2000). The yield loss of 10 per cent is significant as it is sufficient to feed 60 million people for one year. Realizing the importance, natural resource institute of London gave first rank to rice blast disease in its study of pre harvest diseases occurring in South Asia (Gurinder *et al.*, 2006). Several management strategies have been proposed and evaluated to minimize the blast disease incidence. Cultural practices, host plant resistance and the use of synthetic fungicides are the three strategies adopted to control rice

blast (Ghazanfar *et al.*, 2009). Host cultivars, resistant to blast, are the most widely used method of disease control but the host resistance has often failed in field condition, due to the continuously evolving races of *P. oryzae* (Dean *et al.*, 2005). Fungicide application to control the disease is feasible and effective. Hence, the present study was undertaken to evaluate the efficacy of triazole, strobilurin, and their combination fungicides, along with a biocontrol agent, under nursery conditions.

Materials and methods

Ten fungicides belonging to triazoles, strobilurins and their combinations, were tested to study their efficacy in

the management of rice blast in comparison with one bio control agent *P. fluorescens* and control without any spray. The bio efficacies were evaluated against the leaf blast disease in a field laid out in Uniform Blast Nursery (UBN) pattern. The treatments were imposed after noticing the initial disease symptoms. This evaluation was done on highly susceptible rice variety HR-12 in UBN beds (where high disease pressure was maintained) at IIRR, Hyderabad. UBN beds were divided into mini plots with size of 1m × 1m. For each treatment three replications were maintained. Untreated plots were maintained as control. The data were recorded on severity of the infection using Standard Evaluation System (SES) scale (0-9 ratings) (IRRI, 1996), and the Per cent Disease Index (PDI) was calculated using the following formula.

$$\text{PDI} = \frac{\text{Total sum of individual ratings}}{\text{No of samples observed} \times \text{Maximum disease grade}} \times 100$$

No of samples observed × Maximum disease grade

Statistical analysis

The data were subjected to analysis of variance (ANOVA) following the standard procedures described by Gomez and Gomez (1976). Statistical analysis was performed using OPSTAT software (Sheoran *et al.*, 1998). Treatment means were compared using the least significant difference

(LSD) test at $P \leq 0.05$.

Results and discussion

The results indicated that among the treatments, the least Per cent Disease Index (PDI) was observed in Trifloxystrobin + Tebuconazole @ 0.08% (13.33%) followed by Tricyclazole @ 0.06% (14.41%) and Pyraclostrobin @ 0.04% (18.14%). The differences among the treatment were statistically not significant. The PDI of the treatments Azoxystrobin + Tebuconazole @ 0.15%, Azoxystrobin + Difenconazole @ 0.1% and Tricyclazole + Tebuconazole @ 0.1%, Azoxystrobin @ 0.1% (29.57%), Tricyclazole + Tebuconazole @ 0.05%, Tebuconazole @ 0.15% and Difenconazole @ 0.2% ranged from 25.16% to 34.03% among these treatments were statistically at par. The significantly highest PDI of 90.53% was observed in control treatment. *Pseudomonas fluorescens* @ 0.5% registered a PDI of 35.90% which was followed by Difenconazole @ 0.2% (34.03%), Tebuconazole @ 0.15% (32.10%), Azoxystrobin @ 0.1% (29.57%) and Tricyclazole + Tebuconazole @ 0.05% (29.87%) and there was no significant differences among these treatments (Table 1, Figure 1).

Table 1. Bio-efficacy of fungicides and bio control agents against rice leaf blast disease

Sl. No	Group	Fungicide	Mean Per cent Disease Index*	% Reduction over control
1	Strobilurins	Azoxystrobin @ 0.1%	29.57±1.35 (32.91) ^{abcde}	60.96
2	(QoI-fungicides)	Pyraclostrobin @ 0.04%	18.14±1.44 (25.16) ^{fg}	72.39
3	Triazoles	Tricyclazole @ 0.06%	14.41±1.10 (22.28) ^g	76.12
4	(DMI-Fungicides)	Tebuconazole @ 0.15%	32.10±2.39 (34.47) ^{abc}	58.43
5		Difenconazole @ 0.2%	34.03±3.74 (35.62) ^{ab}	56.50
6		Trifloxystrobin+Tebuconazole @ 0.08%	13.33±3.33 (21.13) ^g	77.20
7	Strobilurin+Triazole combination fungicides	Azoxystrobin+Tebuconazole @ 0.15%	25.16±6.40 (29.76) ^{bcdef}	65.37
8		Azoxystrobin+Difenconazole @ 0.1%	25.17±4.63 (29.92) ^{bcdef}	65.36
9		Tricyclazole+Tebuconazole @ 0.05%	29.87±0.58 (33.11) ^{abcd}	60.66
10		Tricyclazole+Tebuconazole @ 0.1%	25.87±4.29 (30.39) ^{bcdef}	64.66
11	Biocontrol agent	<i>Pseudomonas fluorescens</i> @ 0.5%	35.90±2.26 (36.77) ^a	54.63
12		Control	90.53±4.76 (75.22)	
		C.D. (0.05)	8.52	
		SE(m)	2.88	
		SE(d)	4.08	
		C.V.	14.75	

Note: QoI- Quinone outside Inhibitors (Respiration cycle targeted); DMI-DeMethylation Inhibitors (Sterol biosynthesis targeted).

Values in parenthesis are angular transformed values. Values with the similar alphabets in the column are statistically not significant at $P < 0.05$.

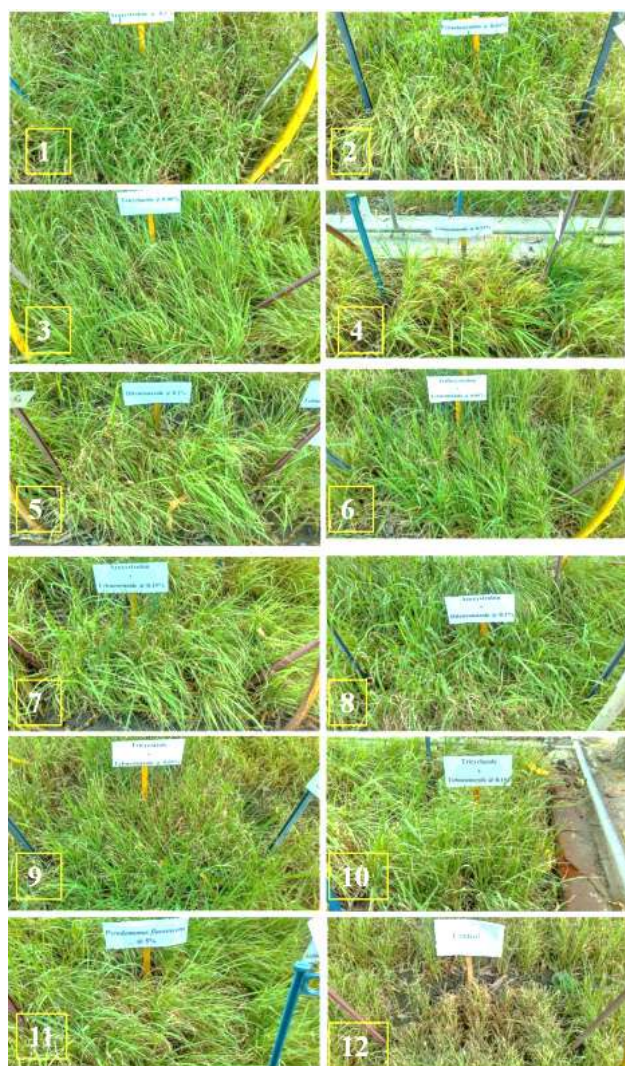


Figure 1. Evaluation of fungicides against rice blast on a susceptible cultivar HR-12 under UBN condition

The results of present study were similar to the findings of Singh and Prasad (2007) who reported tricyclazole (beam) as most effective fungicide for the control of rice blast and increase the yield. Similar result was also reported by Prajapati *et al.* (2004). Effectiveness of Trifloxystrobin 25% + Tebuconazole 50% and Pyraclostrobin 100 g/l CS in controlling rice blast and increasing grain yield was also been observed by Paramesh *et al.* (2016a and b). Vidhyasekaran *et al.* (1997) reported that *P. fluorescens* applied as foliar spray, suppresses rice blast in field condition. The possible reason for reduction of the disease by Tricyclazole, Trifloxystrobin 25% + Tebuconazole 50% and Pyraclostrobin might be due to their long-term inhibition of *P. oryzae* sporulation.

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Sustainable management of mulberry root rot using marine algal bioactive compounds

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Abstract

Seaweeds are the good source of essential nutrients, excellent eco-friendly organic manure and alternative source for inorganic chemical fertilizer for sustaining the crop yield. It is an excellent source of bioactive components with reported antimicrobial activity. The in vitro evaluation of seaweed extract against mulberry root rot has been carried out through poison food technique, agar well diffusion, and disc diffusion assays. After proper inoculation and evaluation indicated that the seaweed extract @ 10000 ppm was found to be best with 71.1 per cent mycelial inhibition and COC @ 1000 ppm was found to be 92.6 per cent mycelial inhibition. From the results of in vitro evaluation, in planta evaluation has been carried out using mulberry seedlings with 3 to 4 leaf stage. The pathogen has been pre inoculated in each seedling and symptoms observed after 24 hours of inoculation. After two days the best treatment has been inoculated through soil drenching and foliar application. It has been found that SE @ 10000 ppm has shown results with disease reduction. Even though the per cent mycelial inhibition was lower than copper oxy chloride, higher concentration may be effective against the disease incidence and safe to the environment.

Keywords: Mulberry root rot, *Macrophomina phaseolina*, Seaweed extract, PFT, AWDA, DDA

Introduction

Mulberry (*Morus* spp.) plant is a perennial fast growing woody plant, which is widely cultivated in many tropical and subtropical areas. The leaves of the mulberry are economically important as they are the only source of feed for the monophagous silkworm (*Bombyx mori* L.) After China, India is one of the leading producers of mulberry with a total acreage of around 2,39,926 hectares. The tropical climate of South India is highly favorable for mulberry cultivation where the crop is harvested at every 70 days (Gnanesh *et al.*, 2021). Mulberry production is limited by several diseases as the perennial nature of the mulberry provides scope for long-term survival and multiplication of soil-borne pathogens (Sharma *et al.*, 2003). Among the soil-borne diseases, root rot is epidemic in nature and a major limiting factor in the cultivation of mulberry. Root

rot disease causes 30% mortality of plants with a 14% decrease in leaf yield, besides deteriorating the leaf quality (Chowdary, 2006; Rajeswari and Angappan, 2018).

In mulberry, different kinds of root rot have been reported, such as black root rot, dry root rot, charcoal root rot, violet root rot, white root rot, *Armillaria* root rot, *Rhizopus* rot, and bacterial root rot (Gnanesh *et al.*, 2021). Molecular phylogeny, identification and pathogenicity of *Rhizopus oryzae* associated with root rot of mulberry in India. (Radhakrishnan *et al.*, 1995; Sharma *et al.*, 2003; Yoshida *et al.*, 2001). Among them, black root rot (BRR) caused by *Lasiodiplodia theobromae* syn. *Botryodiplodia theobromae*, dry root rot (*Fusarium solani* and *F. oxysporum*), and charcoal root rot (*Macrophomina phaseolina*) are frequently reported in India (Pinto *et al.*, 2018).

Lasiodiplodia species are harmful fungal pathogens and have been associated with nearly 500 plant hosts causing various degrees of crop losses. It causes root rot disease in mulberry that affects all parts of the plant and it spreads rapidly affecting a large number of plants in a short period leading to the abandonment of mulberry gardens (Pappachan *et al.*, 2020). The fungus prefers 25 - 30 °C for good growth and survives on all types of substrates. After pruning, the pathogens enter the host through the cut ends of the stem. Once, the plant becomes vulnerable to infection, the fungus dominates inside the roots by multiplying the hyphae rapidly in the cortical tissues and extending up to the pith region. It enters the xylem vessels and causes the death of the plants (Sharma *et al.*, 2003).

Diseased plants appeared stunted with dark-coloured stems and chlorotic leaves, gradually the leaves get wilted and defoliate prematurely. The fine feeder roots and the main root system of these infected plants will damage severely with typical symptoms like discolouration of root xylem followed by the decay of root cortex. also. The bark of the roots also becomes fragile and emanates a foul odour. Gradually as the damaged roots could not firmly hold the plant in the soil they can be easily uprooted (Sowmya *et al.*, 2018; Xie *et al.*, 2014).

The macroscopic marine algae are known as “Seaweeds”. They are used as human food, cattle feed, a chemical fertiliser substitute, and a source of various fine chemicals. Aside from that, it is used to make a variety of agricultural goods like agar and alginate (Khan *et al.*, 2009). Since they contain several growth regulators such as, Cytokinins (Duran *et al.*, 2003), Auxins, Gibberellins and various macro and micronutrients needed for plant growth and development, seaweed extracts are marketed as liquid fertilisers and bio-stimulants. It aids in the production of beneficial soil microorganisms (Khan *et al.*, 2009), develops tolerance against environmental stress (Zhang *et al.*, 2003), increases nutrient uptake from soil (and Groffithi) and enhances antioxidant properties (Verkleij 1992).

These seaweeds consist of 80 - 90% moisture. The dry seaweed extracts contain 50% carbohydrates, protein 10 - 47% with high proportions of essential amino acids, 1-3% lipids and 7-38% minerals. The essential nutrients contain in seaweeds are nitrogen, potassium, phosphorous, calcium, magnesium, sulphur, iron, sodium, zinc, and copper. There are three major classes of seaweed species such as Phaeophyceae, Rhodophyta and Chlorophyta.

Around 700 species of marine algae are present and approximately 60 of them being commercially important in both the intertidal and deep-water regions of the Indian coast. Tamil Nadu, Gujarat, Maharashtra, Goa, Lakshadweep, Andhra Pradesh, and Karnataka states are major seaweed extracts producers. Seaweed acts as a good source of essential nutrients, excellent eco-friendly organic manure and alternative source for inorganic chemical fertilizer for sustaining the crop yield. It is an excellent source of bioactive components with reported antimicrobial activity.

In the present study, the marine algae bioactive compounds were evaluated for efficacy against the mulberry root rot disease.

Materials and methods

Collection of seaweeds

The seaweeds were collected from the coastal region of Manamelkudi, Pudukkottai, Tamil Nadu. The seaweeds were then washed to remove their undesired particles and other sea debris and kept for shade drying for minimum 5-6 days. After complete drying the seaweeds are grinded and made into coarse powders. 8g of powdered seaweeds were taken with 200 ml of distilled water and mixed thoroughly in conical flask. The mixture was then kept in water bath for 2 hours at 80 °C. The seaweed extract was obtained using whatmann filter paper and stored in the amber bottle, kept in refrigerator for further use.

Collection and isolation of the samples

Severely affected mulberry plants show stunted growth with marginal chlorosis, vascular discolouration, and necrosis of leaf with black fungal mycelia on the root epidermis were collected for isolation. The collected

diseased samples with healthy portions were cut into small pieces and the bits were washed in 1% sodium hypochlorite for 60 seconds and subsequently 3 times in sterile water. The bits were transferred into petri dishes containing PDA medium and placed under room temperature. After seven days of inoculation the pathogen completed its full growth in a petri dish of 9 cm. Microscopic observations were made for the pathogen identification.

***In vitro* evaluation of seaweed extract against mulberry root rot**

In vitro evaluation of seaweed extract against mulberry root rot was carried out with seven treatments and three replications using the CRD experimental design. *In vitro* evaluation carried out by two assays viz poisoned food technique, agar well diffusion assay, and disc diffusion assay.

In poisoned food technique (Table 1), double stranded potato dextrose agar medium has been used for pathogen inoculation. To avoid bacterial contamination the seaweed extract was filtered in 0.2 mm bacterial proof filter. The filtered extract was then pipetted out according to the given concentrations and mixed well in the medium before pouring into the petri plates. After solidification of the medium the pathogen was inoculated into the petri plates and kept under room temperature.

In agar well diffusion assay (Table 2), double stranded potato dextrose agar medium was used for pathogen inoculation. To avoid bacterial contamination the seaweed extract was filtered in 0.2 mm bacterial proof filter. The filtered extract was pipetted out according to the given concentrations. After solidification of the medium, four wells were made by using corn borer, and the filtered extract at different concentrations was poured into each disc. Thereafter the pathogen was inoculated into each plate and kept under room temperature.

In disc diffusion assay (Table 3) after solidification of the potato dextrose medium, four bits of filter paper which has been immersed in different concentrations of seaweed extract was kept into each petri dish. Then the pathogen has been inoculated into each petri dish and kept under room temperature.

***In planta* evaluation of seaweed extract against mulberry root rot**

The best concentration of seaweed obtained from the *in vitro* studies was used for the *in planta* evaluation and the per cent disease index was be calculated. CRD designs was used with six treatments and four replications. The variety used is V1. For *in planta* evaluation, 24 semi wood mulberry cuttings were collected and planted in the polybags containing red sand, FYM, and native soil which was irrigated continuously for the growth of the plant.

For pathogen inoculation in plants PD broth was prepared and autoclaved. The mycelial disc of the pathogen was inoculated for the conidial production (1×10^6 / ml). The conidial count was confirmed by using haemocytometer and inoculated into each plant. After three days, the symptoms were identified and treatments were applied (Table 4.). The best concentration obtained from *in vitro* evaluation as 1000 ppm.

Results and discussion

Collection of phaeophycean seaweed

The seaweeds were identified as marine brown algae (Class. Phaeophyceae) based on the appearance of olive yellow to deep brown in colour due to the presence of fucoxanthin. A short main axis, alternatively organised primary cylindrical branches with branchlets, around 30-40 cm in height, and vesicles that were spherical, pointed, stalked, and ovate about 3-4mm wide, and it grows on rocks and has a discoid holdfast observed and confirmed as *Sargassum tennerimum* (J. Agardh) (Fig. 1).



Figure 1. *S. tennerimum*

Identification of Pathogen

Based on the cultural characters such as greyish black mycelium under *in vitro*, morphological characters such as septate mycelium with flame shaped conidia the pathogen was confirmed as *Macrophomina phaseolina* (Fig. 2). The pathogenicity was proven by adopting Koch's postulates.



Figure 2. Symptoms, cultural, microscopical and pathogenicity of *M. phaseolina*

In vitro evaluation of seaweed extract against mulberry root rot

After seven days of inoculation, in the poison food technique, the standard check at the concentration of 1000 ppm inhibited maximum of 59.25% of mycelial growth followed by the SE @ 10000 ppm exhibited 41.6% of mycelial inhibition. There is no mycelial inhibition observed from the treatment T₁ (Table 1).

Table 1. Per cent inhibition of mycelium under poisoned food technique

Treat-ment	Treatment details	Radial growth of mycelium (in cm)	Percentage of mycelial inhibition
T1	SE @ 500 ppm	9	0(0.000) ^d
T2	SE @ 1000 ppm	8.75	2.8(2.77) ^d

T3	SE @ 2500 ppm	7.3	18.85(18.88) ^c
T4	SE @ 5000 ppm	5.76	35.8(35.92) ^b
T5	SE @ 10000 ppm	5.25	41.6(41.66) ^b
T6	COC @ 1000 ppm	3.66	59.2(59.25) ^a
T7	Control	9	0(0.00) ^d
		CD (0.05)	1.43
		SE (mean)	2.12

*Mean of three replications, Observations were taken after 7 days of inoculation

Figures with same letter do not have significant difference according to one way ANOVA at P=0.05

In agar well diffusion assay, after seven days of inoculation, the concentration of 1000 ppm inhibited maximum of 88.8% of mycelial growth followed by the SE @ 10000 ppm showing 62.2% of mycelial inhibition. There was no mycelial inhibition observed from the treatment T₁ (Table 2.).

Table 2. Per cent inhibition of mycelium under agar well diffusion assay

Treat-ment	Treatment details	Radial growth of mycelium (in cm)	Percentage of mycelial inhibition
T1	SE @ 500 ppm	9	0(0.000) ^d
T2	SE @ 1000 ppm	8.73	1.85(2.96) ^d
T3	SE @ 2500 ppm	6.46	28.1(28.14) ^c
T4	SE @ 5000 ppm	3.3	63.3(63.33) ^b
T5	SE @ 10000 ppm	2.6	62.19(62.21) ^b
T6	COC @ 1000 ppm	1	88.8(88.54) ^a
T7	Control	9	0(0.00) ^d
		CD (0.05)	2.4
		SE (mean)	3.12

*Mean of three replications, Observations were taken after 7 days of inoculation

Figures with same letter do not have significant difference according to one way ANOVA at P=0.05

In the disc diffusion assay, after seven days of inoculation, the concentration of 1000 ppm inhibited maximum of 92.6% of mycelial growth followed by the SE @ 10000 ppm exhibiting 71.1% of mycelial inhibition. There was no mycelial inhibition observed in the treatment T₁ (Table 3.).

Table 3. Per cent inhibition of mycelium under disc diffusion assay

Treat-ment	Treatment details	Radial growth of mycelium (in cm)	Percentage of mycelial inhibition ^d
T1	SE @ 500 ppm	9	0(0.000) ^d
T2	SE @ 1000 ppm	7.6	15.55(15.55) ^c
T3	SE @ 2500 ppm	6.41	28.7 (28.7) ^d
T4	SE @ 5000 ppm	4	55.5 (55.55) ^c
T5	SE @ 10000 ppm	3.4	71.1 (67.03) ^b
T6	COC @ 1000 ppm	0.66	92.6 (92.58) ^a
T7	Control	9	0 (0.00) ^d
		CD (0.05)	2.31
		SE (mean)	1.95

*Mean of three replications, Observations were taken after 7 days of inoculation

Figures with same letter do not have significant difference according to one way ANOVA at P=0.05

In the evaluation of study on lesion size in T5, the symptoms were observed as severe and disease score was related as 9 while, healthy plants of T6 showed no symptoms. The confirmed application methods of seaweed extract exhibited maximum pathogen control and produced few lesions on par with the standard check (Table 3)

In planta evaluation of seaweed extract against mulberry root rot

In treatment 5, the symptoms have been observed as severe and rated disease score as 9, and healthy plants produces no symptoms (T₆). The combined application methods of seaweed extract exhibit maximum of pathogen control and produces few lesions which is on par with the standard check. (Table 5.).

Table 5. Evaluation of lesion size after the treatment of seaweed extract

Treatment	Treatment details	Disease scoring
T1	SE (Soil Drenching)	5-6 (Moderate)
T2	SE (Foliar Application)	6-8 (Many lesions)
T3	SE (Soil Drenching + Foliar Application)	1-3 (few lesions)
T4	COC (Soil Drenching)	1-3 (few lesions)
T5	Pathogen alone	9-Severe infection
T6	Healthy control	1 - No infection

The antimicrobial property of the seaweed against *M. phaseolina* and the potential of mycelial inhibition was shown through *in vitro* analysis (Figs. 1, 2 and 3). Several scientific studies *in vitro* revealed the antimicrobial potential of seaweeds. Application of dried powder forms of *S. tenerrimum*, *S. wightii*, *S. swartzii* as a soil amendment in *Helianthus annuus*, effectively decreased the disease incidence of root rot caused by *M. phaseolina* and *F. solani* in plants (Ara *et al.*, 1996).

In comparison with ethanol, ethyl acetate, and water extracts (Thinakumar and Sivakumar, 2013) *S. wightii* and *S. lilicifolium* extracts in petroleum ether, benzene, and chloroform caused the most significant decrease in *P. aphanidermatum* mycelial growth. Ambika and Sujatha (2014) reported that *Sargassum myricocystem* demonstrated significant antifungal activity against *Fusarium oxysporum* f. sp. *udum* by employing the poisoned food method with the mycelial inhibition ranges at various intervals in redgram.

Hernández-Herrera *et al.* (2014) reported green algae extracts (*Ulva lactuca* and *Caulerpa sertularioides*), and brown algae extracts (*Padina gymnospora* and *Sargassum liebmannii*) provided protection against the necrotrophic fungus *Alternaria solani* in tomato plants (*Solanum lycopersicum* L.) through soil drenching and foliar application. The application of extract derived from brown sea algae (*Sargassum fusiforme*) and its ability to induce resistance in tomato plants significantly reduced the severity of late blight, grey mold, and powdery mildew in tomatoes (Sbaihat *et al.*, 2015).

A lower concentration of 15 (%) *Turbinaria conoides* extract effectively inhibits the mycelial growth of *Fusarium oxysporum* fungi, and the GC-MS analysis shows the presence of several antimicrobial compounds may be inducing the antifungal activity (Selvaraju and Vijayakumar, 2016).

The inhibitory activity of aqueous and organic extracts of *Sargassum vulgare* from Tunisian coastal areas against the potato leak-causing agent, *P. aphanidermatum*, was tested *in vitro* and *in vivo*, resulting in suppressing the disease

severity. The activity was modified by the types of extracts examined, sample locations, and concentrations used (Ammar et al., 2017).

The application of the commercial extract ((Algamar®) *A. nodosum* at 0.4 mL L⁻¹ significantly decreased the incidence and severity of brown rot in the infected plum fruits and exhibited a similar control effect as the fungicide iprodione (Viencz et al., 2020).

Overall the present study showed the effectiveness of seaweed extracts involving marine algal bioactive compounds for the management of mulberry root rot.

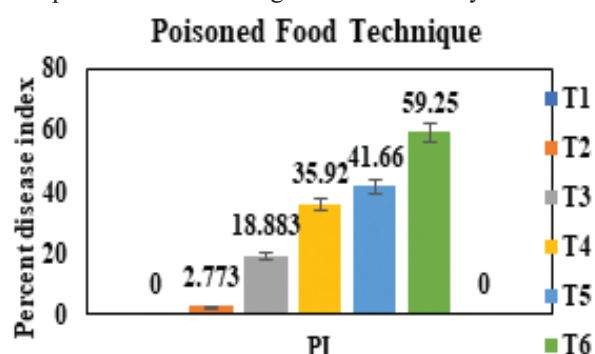


Fig. 3. Per cent of mycelial inhibition of poisoned food technique

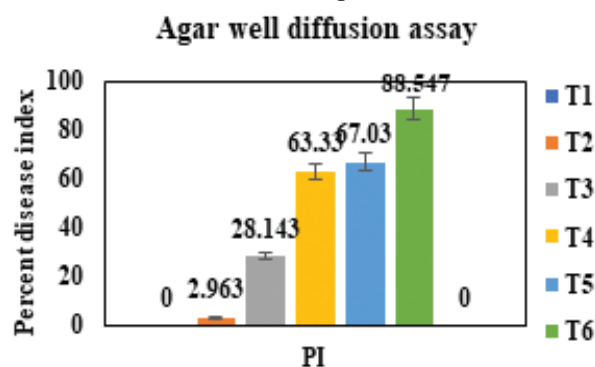


Fig. 4. Per cent of mycelial inhibition of Agar Well Diffusion Assay

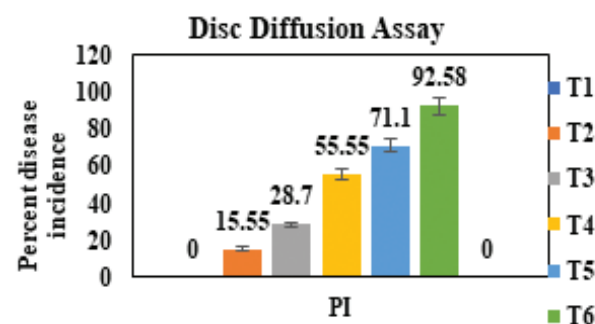


Fig. 5. Per cent of mycelial inhibition of Disc Diffusion Assay

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Field efficacy of new insecticide molecules against *Spodoptera frugiperda* (J.E Smith) and dissipation studies

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Abstract

*A field trial of new insecticides was carried out against fall armyworm in maize during rabi, 2019-20 and 2020-21 at the College farm and studies on dissipation of selected insecticides were conducted at Pesticide Residues Laboratory, AINP on Pesticide Residues, PJTAU, Rajendranagar, Hyderabad. The bio-efficacy studies revealed that spinetoram was the most effective insecticide against *S. frugiperda* with 97.61 per cent reduction in population followed by indoxacarb (91.09%), spinosad (71.43%), emamectin benzoate (60.21%) and chlorantraniliprole (55.56%). The studies on dissipation pattern of selected insecticides in immature maize cobs revealed that indoxacarb persisted for 1.05 days while spinetoram dissipated below the limit of quantification (10 ppb), 2hrs after spraying.*

Keywords: *S. frugiperda*, new insecticides, spinetoram, indoxacarb and spinosad.

Introduction

Fall armyworm (FAW) (*Spodoptera frugiperda*) (J.E Smith) (Noctuidae: Lepidoptera) was reported in Africa during 2016, followed by an alarming distribution across the African continent. It was reported in India during 2018 and several Asian Pacific countries; Bangladesh, Nepal, Myanmar, Thailand, and China. Fall armyworm poses a threat outside its range particularly in temperate regions because adult armyworm can travel several hundred kilometres in a single by flying to and maintaining an elevation of several hundred meters, at which height winds can transport them in a directional manner (Westbrook *et al.*, 2016).

The FAW larvae generally feed by remaining in the whorl of maize for most of its life thus reducing its contact with insecticides (FAO, 2018), and necessitating spraying of insecticides multiple times. Repeated application of insecticides may lead to the quick development of resistance (Gutierrez-Moreno *et al.*, 2019). Hence, to

delay the development of resistance to insecticides, it is essential to rotate the chemicals having different modes of action. At present, the Central Insecticide Board and Registration Committee (CIB & RC) has recommended the use of chlorantraniliprole 18.5 SC, thiamethoxam 12.6% + lambda cyhalothrin 9.5% ZC, and spinetoram 11.7 SC (DPPQS, 2019) for fall armyworm management. In the current research eight new insecticide molecules with diverse modes of action were screened against FAW in field condition for two seasons in maize crop which would be one of the helpful components of integrated management of this insect pest in future.

With the sudden ingress of the most devouring herbivore FAW into our nation, there was no time to decide the chemical management. So, the problem of pesticide residues has arisen because of lack of knowledge among farmers with respect to appropriate use of chemicals against this pest. The pesticide residues may enter edible products

via indiscriminate use of insecticides and disregard to the given waiting periods. Therefore, consumption of pesticide residues through maize products such as popcorn and maize grains is a possibility. Farmers sometimes use cocktail of two or more chemicals which cause more hazards to the environment and public health. Throughout the processing of maize grains like polishing and milling washing with water is not done anywhere. These evidences enthused us to conduct a study on pesticide residues in maize at different intervals after application of insecticidal spray by using LC-MS/MS. This method recognized as ideal, highly specific, and sensitive for testing agricultural commodities. Therefore, in the present study the levels of pesticide residues, dissipation percentage and half-life in raw maize cobs collected from experimental trial were determined at different intervals after spraying.

Materials and methods

Field screening of new insecticide molecules on maize

Table 1. Details of insecticide treatments

Treatment No.	Technical name	Trade name	Dosage (g. a.i/ha)	Dosage (ml/l)	Class
T ₁	Indoxacarb	King doxa 14.5% SC	72.5	1	Oxadiazine
T ₂	Chlorantraniliprole	Coragen 18.5% SC	37	0.4	Anthranilic Diamide
T ₃	Emamectin benzoate	Proclaim 5% SG	10	0.4	Avermectin
T ₄	Spinetoram	Delegate 11.7% SC	58.5	1	Spinosyn
T ₅	Novaluron	Rimon 10% EC	50	1	Benzoyl urea
T ₆	Spinosad	Tracer 45% SC	90	0.4	Spinosyn
T ₇	Chlorantraniliprole+ Lambda cyhalothrin	Ampligo 9.3% + 4.6% ZC	27.8	0.4	Pyrethroids & diamides
T ₈	Flubendiamide	Fame 39.35% SC	72	0.3	Anthranilic diamide
T ₉	Untreated control	-	-	-	-

Insecticidal Application

Test insecticides were applied using a high-volume knapsack compression sprayer when pest infestation reached ETL. The ETL (Economic Threshold Level) for Fall army worm are as follows: a) 1-2 Larvae per whorl; b) 5% Seedling are cut; c) 15% whorl infestation of young plants during first 30 days. The spraying was taken up during morning hours to prevent the drift of spray fluid reaching the adjacent plots.

Data Collection

The observations on pest incidence and their numbers were recorded one day before spraying as pre-treatment count. Post treatment count was recorded at three, five and seven

against fall armyworm was carried out during *rabi*, 2019-20 and *rabi*, 2020-21 at the College farm, situated at an altitude of 542.3 m above mean sea level with 17°19'N latitude and 78°25' E longitude. The laboratory analysis to study the dissipation pattern of insecticides was undertaken at Pesticide Residues Laboratory, AINP on Pesticide Residues, PJTAU, Rajendranagar, Hyderabad.

To evaluate the efficacy of new insecticide molecules against *S. frugiperda*

To explore the effectiveness of insecticides with diverse mode of action against transboundary pest, *S. frugiperda* infesting maize, the field experiment was carried out in Randomized Block Design (RBD) with nine treatments and three replications during *rabi* 2019-20 and 2020-21 (Table 1). The maize hybrid Pioneer- 3396 was sown. All the agronomical procedures performed in the field were as per PJTAU recommended agricultural practices.

days after spraying. For recording the pest population counts, five plants were randomly selected in each plot, excluding border rows. The data on cob damage was taken at the time of harvest through counting of total number of cobs and number of damaged cobs, plot wise.

The per cent reduction in different treatments over control was calculated by modified Abbott's formula (1987) as given below.

X : per cent living in the check.

Y : per cent living in the treated plot.

X - Y : per cent killed by the treatment.

And the per cent killed by the treatment (X - Y) divided by the per cent living in the check (X) expressed by an equation,

$$\frac{X-Y}{X} \times 100$$

Statistical analysis

The data from various experimental plots were analyzed following appropriate statistical methods (RBD) suggested by Gomez and Gomez (1984). Data were transformed using square root transformation or angular transformation as required before statistical analysis.

Results and discussion

Efficacy of Insecticides against Fall Armyworm during *rabi*, 2019-20.

The population of fall armyworm in various treatments a day before first spraying ranged from 1.06 to 1.93 larvae/plant (Table 2). There was no significant difference in population among the various treatments, indicating homogenous distribution of larvae before spraying.

The mean larval population of fall armyworm at 3 DAS indicated that all the insecticidal treatments were significantly superior over control (1.93 larvae/plant). The population was significantly lowest (0.40 larva/plant) in plots sprayed with spinetoram on par with indoxacarb (0.46 larva/plant), spinosad (0.60 larva/plant), emamectin benzoate (0.73 larva/plant) and chlorantraniliprole (0.86 larva/plant). The next best treatments in managing fall armyworm were novaluron (0.93 larva/plant) on par with followed by chlorantraniliprole + lambda cyhalothrin (1.00 larvae/plant) and flubendiamide (1.13 larvae/plant).

At 5 days after first spray, spinetoram treatment continued to be effective by recording significantly lowest fall armyworm population of 0.33 larva/plant. It was found to be on par with indoxacarb (0.40 larva/plant) and spinosad (0.40 larva/plant) followed by emamectin benzoate, novaluron and chlorantraniliprole with 0.60, 0.73 and 0.73 larva per plant, respectively, which were found to be on par. However, flubendiamide and chlorantraniliprole + lambda cyhalothrin treated plots recorded highest population of 0.80 and 1.00 larva/plant on par with control (1.33 larvae/plant).

At 7 days after first spray, spinetoram, indoxacarb, spinosad and emamectin benzoate with 0.06, 0.13, 0.20 and 0.53 larva per plant were at par and significantly superior to control (1.26 larvae/plant) but were on par with each other. They were followed by chlorantraniliprole (0.60 larva/plant), novaluron (0.66 larva/plant), flubendiamide (0.73 larva/plant) and chlorantraniliprole + lambda cyhalothrin (0.93 larva/plant) treatments which were found to be less effective but had significantly lower population than control.

The per cent reduction of fall armyworm in different treatments over control in decreasing order of efficacy was spinetoram, indoxacarb, spinosad, emamectin benzoate, chlorantraniliprole, novaluron, flubendiamide and chlorantraniliprole + lambda cyhalothrin with 95.23, 89.68, 84.12, 57.93, 52.38, 47.61, 42.06 and 26.19 per cent reduction, respectively.

During 2019-20, the population of fall armyworm after I spray was very low in the treatments and hence the need for spray -II was not required.

Rabi, 2020-21

The result revealed that the pre count of fall armyworm, *S. frugiperda* in different treatments ranged between 1.00 to 1.26 larvae/plant (Table 2) and was non-significant showing even distribution of plant infestation before spraying.

The mean larval population of fall armyworm at 3 DAS indicated that all the insecticidal treatments were significantly superior over control (0.93 larva/plant). The population was significantly lowest (0.06 larva/plant) in plots sprayed with spinetoram and found to be superior over other treatments, followed by indoxacarb (0.40 larva/plant), chlorantraniliprole + lambda cyhalothrin (0.46 larva/plant), emamectin benzoate (0.53 larva/plant), chlorantraniliprole (0.53 larva/plant) novaluron (0.60 larva/plant) and spinosad (0.60 larva/plant) which were on par with each other. However, maximum number of larvae among the insecticidal treatments was noticed in flubendiamide (0.86 larva/plant).

Observations at 5 DAS indicated that spinetoram was significantly superior to other treatments with nil larval population followed by indoxacarb, emamectin benzoate, chlorantraniliprole, chlorantraniliprole + lambda

cyhalothrin, spinosad and novaluron which were on par with the larval population of 0.20, 0.33, 0.40, 0.40, 0.46 and 0.53 larva/ plant, respectively. However, flubendiamide recorded highest population of 0.73 larva/plant, on par with control (0.80 larva/plant). Similarly, spinetoram and indoxacarb were found significantly superior over other treatments at 7 days of treatment with the larval population of 0.00 and 0.06 larva/plant respectively, although indoxacarb was found to be at par with emamectin benzoate. Emamectin benzoate, chlorantraniliprole, spinosad, chlorantraniliprole + lambda cyhalothrin and novaluron were on par with 0.26, 0.33, 0.33, 0.40 and 0.40 larva/ plant, respectively, while the population in the control was 0.80 larva/plant. The per cent reduction over control was highest in spinetoram (100.00%) followed by indoxacarb (92.50%), emamectin benzoate (62.50%), chlorantraniliprole (58.75%) spinosad (58.75%), novaluron (50.00%), chlorantraniliprole+lambda cyhalothrin (50.00%) with the lowest being recorded in flubendiamide treated plot (25.00%).

The overall mean per cent reduction was highest in plots treated with spinetoram (97.61%) followed by indoxacarb (91.09%), spinosad (71.43%), emamectin benzoate (60.21%), novaluron (48.80%), chlorantraniliprole+lambda cyhalothrin (38.09%) and lowest in flubendiamide (33.53%) (Fig. 1.).

From the present study it is evident that spinetoram was the most effective insecticide against the transboundary pest, fall armyworm. The results corroborate with the reports of Bonni *et al.* (2020). who recorded that spinetoram was the most effective insecticide with a reduction rate of 70.77% of fall armyworm in maize plants, followed by emamectin benzoate (54.86%) and azadirachtin (36.36%). Also, Thumar *et al.* (2020). reported that spinetoram, emamectin benzoate, chlorantraniliprole and thiodicarb, were found more effective in checking the larval population of fall armyworm. Mallapur *et al.* (2019) observed that spinetoram, emamectin benzoate and spinosad were significantly superior over all other treatments with the larval reduction of 98.13, 96.26 and 96.26 per cent, respectively at 7 days after treatment imposition. Similarly, Dileep Kumar and Murali Mohan (2020) assessed that spinetoram was highly effective in reducing the larval population of fall armyworm and leaf

damage, followed by novaluron and chlorantraniliprole with 97.32, 93.09 and 90.43 per cent reduction in larval population, respectively over untreated control. In Ethiopia, fall armyworm was effectively controlled using insecticides *viz.*, spinetoram, chlorantraniliprole, spinosad and lambda cyhalothrin (Sisay *et al.*, 2019). Whereas, Kumar and Muthukrishnan (2017) observed 69.00 to 84.40 per cent reduction of *Spodoptera litura* Linn. using spinetoram at varying doses in pigeon pea. Visnupriya and Muthukrishnan (2017) observed 72.4 to 83.9 per cent reduction of *Helicoverpa armigera* (Hub.) in okra. Spinetoram is a non-systemic insecticide derived from active biological substances (spinosyn) produced by the soil actinomycete *Saccharopolyspora spinosa*. It is semi-synthetic molecule discovered during studies of modification of natural substances by Dow Agro Sciences. Spinetoram has a broad spectrum of action and can be used on several crops. It is applied in low doses and has a low impact on most beneficial insects (Mertz and Yao, 1990). It is nicotinic acetylcholine receptor (nAChR) allosteric modulator- site-I.

The next best treatment for the control of fall armyworm was indoxacarb. Current results coincide with the field efficacy studies of Kumari *et al.* (2020) who also reported the superior efficacy of indoxacarb in maize. Indoxacarb is a broad-spectrum oxadiazine group of insecticide effective against lepidopteran pests and selected sucking pests. It has a novel mode of action that blocks the movement of sodium ions into certain nerve cell ion channels, resulting in paralysis and death of pests (Lapied *et al.*, 2001).

The other effective treatments against fall armyworm in the present study were spinosad, emamectin benzoate, chlorantraniliprole. Bajracharya *et al.* (2020) concluded that spinosad, chlorantraniliprole and emamectin benzoate were promising for control of fall armyworm in maize. Whereas, the results of Deshmukh *et al.* (2020) on field efficacy revealed that the effective insecticides were chlorantraniliprole, followed by emamectin benzoate, spinetoram, flubendiamide, indoxacarb, lambda cyhalothrin and novaluron against fall armyworm. Sangle *et al.* (2020) reported that emamectin benzoate to be most effective treatment over control followed by chlorantraniliprole, flubendiamide, indoxacarb, thiamethoxam + lambda cyhalothrin, spinosad and

profenophos against FAW. Susanto *et al.* (2021) reported highest mortality (>80%) with emamectin benzoate, chlorfenapyr, phoxim, methomyl and indoxacarb under laboratory, screenhouse and field conditions against fall armyworm.

Emamectin benzoate is a 16-membered macrocyclic lactone produced by the fermentation of the soil actinomycete *Streptomyces avermitilis* with a strong activity against lepidoptera as well as with a high selectivity on useful organisms. This molecule acts if swallowed and has some contact action. It penetrates leaf tissues (translaminar activity) and forms a reservoir within the leaf. The mechanism of action is unique as it inhibits muscle contraction, causing a continuous flow of chlorine ions in the GABA and H-Glutamate receptor sites (Fanigliulo and Sacchetti, 2008). On other hand, chlorantraniliprole is a novel anthranilic diamide insecticide that functions via activation of the insect ryanodine receptors within the sarcoplasmic reticulum causing impaired regulation of muscle contraction. Ryanodine receptor channels regulate the release of internal calcium stores and are important in muscle contraction. Sustained release of calcium levels within the cytosol leads to muscle contraction, paralysis, and eventual death of the organism. While insects possess a single form of the ryanodine receptor distributed in muscle and neuronal tissue, mammals possess three forms which are widely distributed in muscle and non muscle tissues (USEPA, 2007).

Flubendiamide, chlorantraniliprole+lambda cyhalothrin and novaluron were found to be least effective insecticides for management of fall armyworm in the present study. However, Deshmukh *et al.* (2020) revealed them to be most effective insecticides against fall armyworm.

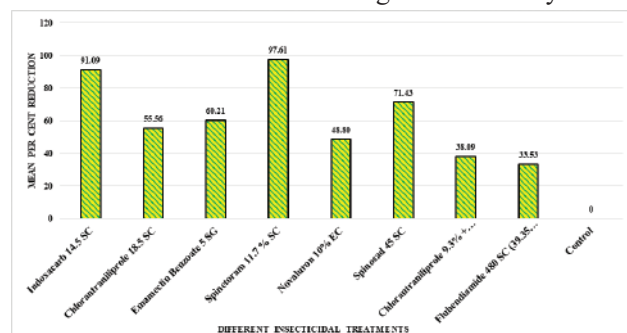


Figure 1. Mean per cent reduction of rabi 2019-20 and 2020-21

Dissipation studies

Dissipation pattern of the two most effective insecticides viz., indoxacarb and spinetoram were studied at 0,1,5,7,10,15, and 25 days after spraying on immature cobs. The method adopted for analysis of the insecticides was LC-MS/MS and evaluated through linearity, LOD, LOQ, fortification, recovery, repeatability and reproducibility studies.

Linearity, Recovery and Dissipation of Indoxacarb

The linearity curve of indoxacarb is depicted in Fig. 1. The correlation coefficient (R^2) value obtained from the linearity curve was 0.992.

The accuracy of indoxacarb as per cent in the immature cob sample fortified at 0.01 ppm was recorded as 100.37 per cent, while at 0.1ppm fortification level, the recovery ranged from 97.40 to 96.66 with an average of 96.59 per cent recovery.

The repeatability at 0.01 and 0.1 ppm fortification levels was 96.83 and 97.03 per cent recovery respectively. Reproducibility at 0.01 and 0.1 ppm fortification levels was 101.49 and 96.35, respectively. As the recoveries were in the acceptable range of 70-120 (SANTE, 2021), the method was suitable for analysis of indoxacarb residues upto 10 ppm and it was taken as limit of quantification (LOQ) (Table 3).

The dissipation pattern of indoxacarb 14.5 SC @ 72.5 g a.i./ha in maize is presented in Table 4 and Fig. 2. The mean initial deposit of 0.029 ppm recorded at 0 days (2 hrs) after spray dissipated to 0.015 ppm at 1 day after spray and to < LOQ from 3rd day onwards. Thus, the per cent dissipation at 1,3 after spray was 48.27, 100.00, respectively. The regression equation obtained $y = -0.2863x + 1.7487$ with R^2 of 1.00. The calculated half-life was 1.05 days.

Takkar *et al.* (2011) reported that the average initial deposits were 0.23 and 0.45 mg kg⁻¹ in cauliflower after last application of indoxacarb @ 52.2 and 104.4 g. a.i. ha⁻¹, respectively at recommended and double the recommended dosages, respectively. The residues dissipated below its LOQ of 0.01 mg kg⁻¹ after 7 days and its half-life periods were observed to be 1.12 and 1.31 days, respectively, at single and double the dosages.

Naik *et al.* (2019) reported that the initial deposits on green pods of pigeon pea following two applications of indoxacarb 14.5% SC at 60 and 120 g a.i. ha⁻¹ was 0.583 and 1.342 µg g⁻¹ and dissipated to 0.005 and 0.006 µg g⁻¹ on 7 and 10 days after application, respectively. Half-life and safe waiting period (SWP) ranged from 1.13 to 1.23 days and 3.23 to 5.03 days for the dose 60 and 120 g a.i. ha⁻¹, respectively in pigeonpea green pods. Pandey *et al.* (2019) reported that the dissipation in okra was maximum on 10th day in emamectin benzoate (99.97%) followed by bifenthrin on 10th day (97.73%), indoxacarb on 7th day (96.66%), deltamethrin on 14th day (96.15%), flubendiamide on 10th day (95.00%) and lambdacyhalothrin on 10th day (89.79%). The half-life (t_{1/2}) values for the above insecticides were 0.6, 2.18, 1.32 to 1.58, 5.9 to 17, 5.5 and 5.0 days, respectively.

The ability to resist degradation (persistence) under various conditions is measured as half-life of the pesticide (Bajwa and Sandhu, 2014). Half-life of a pesticide can range from hours or days to years for more persistent ones. Pesticides can be degraded by photolysis, hydrolysis, oxidation and reduction, metabolism (plants, animals, microbes), temperature and pH (Helfrich, 2009).

Linearity, Recovery and Dissipation of Spinetoram

The linearity curve of spinetoram is depicted in (Figs. 3 and 4). The correlation coefficient (R²) value obtained from the linearity curve was 0.999 and 0.999 for spinosyn -J and L, respectively.

The accuracy of spinetoram (spinosyn-J) in terms of per cent recovery in the immature cob samples fortified at 0.0075 ppm and 0.075 ppm were recorded as 99.01 and 97.55 respectively, whereas, the repeatability (per cent recovery) at 0.0075 ppm and 0.075 fortification levels were 93.64 and 91.83, respectively. Reproducibility (per cent recovery) at 0.0075 and 0.075 ppm fortification levels were 107.36 and 98.38 per cent respectively.

The accuracy of spinetoram (spinosyn-L) in terms of per cent recovery in the immature cob samples fortified at 0.0025 ppm and 0.075 ppm were recorded as 93.89 and 99.60, respectively whereas, the repeatability (per cent recovery) at 0.0025 ppm and 0.075

fortification levels were 87.90 and 100.48, respectively (Tables 5 and 6). Reproducibility (per cent recovery) at 0.0025 and 0.075 ppm fortification levels were 95.99 and 103.11, respectively.

As per SANTE, (2021) guidelines, the recoveries were in the acceptable range of 70-120 per cent. Thus, the method was suitable for analysis of spinetoram residues upto 10 ppm and hence was taken as limit of quantification (LOQ). The dissipation pattern of spinetoram 11.7 SC applied at 58.5 g a.i./ha, showed no residues from '0' day (2 hrs) after spray indicating that both the metabolites of spinetoram *viz.*, J and L degraded rapidly (Table 7).

Thus, from the dissipation studies it can be concluded that both insecticides *i.e.*, indoxacarb and spinetoram can be used safely for management of FAW in maize crop as no residues were detected at harvesting stage (25 days after spraying). According to Lin *et al.* spinetoram swiftly degraded in cauliflower, with half-life of less than or equal to 4.85 days. The terminal residues of total spinetoram (sum of spinetoram and its two metabolites) detected in cauliflower samples were in the range of 0.009-0.337 mg/kg. Malhat showed that spinetoram dissipation pattern followed the first order kinetics with half-life of 2.6 days, in tomato. The spinetoram residues in tomato were below the codex maximum residue level (0.06 mg kg⁻¹) after 10 days of application. Liu *et al.* (2011) reported the levels of residues in two crops (Garland chrysanthemum and Aster scaber) treated under a fixed schedule in the greenhouse as 6.21-0.55 ppm (maximum residue limit = 7 ppm).

The dissipation/degradation of the pesticide residues in crops depends on various factors, including environmental condition (Gennari, 1985), chemical formulation, type of application, plant species, dosage (Cabras, 1989), interval between application, the relation between the treated surface and its weight, and living state of the plant surface, in addition to harvest time (Khay *et al.*, 2008; Cabras 1990). Different half-lives of spinetoram between sites may be affected by a variety of factors, such as weathering, heat decomposition, humidity, and the growth dilution factor (Lin 2019; Malhat *et al.*, 2014), among which growth dilution factor might play a more important role (Malhat *et al.*, 2014).

Table 1.2 Bio-efficacy of insecticides against fall armyworm in maize during *rabi*, 2019-20 and 2020-21.

Treatments	Dose (ml ha ⁻¹)	Number of fall armyworm larvae/plant*									Overall Mean Per cent reduction Spray I&II		
		Rabi, 2019-20					Rabi, 2020-21						
		Pre- Spray	3 DAS I	5 DAS I	7 DAS I	Per cent reduction	Pre-Spray	3 DAS I	5 DAS I	7 DAS I		Per cent reduction	
T ₁	Indoxacarb 14.5 SC	500 ml	1.60 (1.25)	0.46 (0.68) ^{cd}	0.40 (0.61) ^{cd}	0.13 (0.79) ^{de}	89.68	1.06 (1.02)	0.40 (0.94) ^c	0.20 (0.83) ^d	0.06 (0.75) ^{de}	92.50	91.09
T ₂	Chlorantraniliprole 18.5 SC	200 ml	1.33 (1.14)	0.86 (0.90) ^{bcd}	0.73 (0.81) ^{bcd}	0.60 (1.03) ^{bc}	52.38	1.13 (1.06)	0.53 (1.01) ^{bc}	0.40 (0.94) ^{cd}	0.33 (0.91) ^{bc}	58.75	55.56
T ₃	Emamectin Benzoate 5 SG	200 g	1.20 (1.09)	0.73 (0.85) ^{bcd}	0.60 (0.74) ^{bcd}	0.53 (1.00) ^{bcd}	57.93	1.20 (1.08)	0.53 (1.00) ^{bc}	0.33 (0.91) ^{cd}	0.26 (0.87) ^{cd}	62.50	60.21
T ₄	Spinetoram 11.7% SC	500 ml	2.06 (1.40)	0.40 (0.61) ^d	0.33 (0.57) ^d	0.06 (0.75) ^d	95.23	1.26 (1.12)	0.06 (0.75) ^d	0.00 (0.70) ^e	0.00 (0.70) ^e	100.00	97.61
T ₅	Novaluron 10% EC	500 ml	1.06 (1.02)	0.93 (0.96) ^{bc}	0.73 (0.85) ^{abcd}	0.66 (1.07) ^b	47.61	1.13 (1.06)	0.60 (1.04) ^{abc}	0.53 (1.01) ^{bc}	0.40 (0.93) ^{bc}	50.00	48.80
T ₆	Spinosad 45 SC	200 ml	1.33 (1.14)	0.60 (0.76) ^{bcd}	0.40 (0.63) ^{cd}	0.20 (0.83) ^{cde}	84.12	1.00 (0.99)	0.60 (1.03) ^{abc}	0.46 (0.97) ^c	0.33 (0.91) ^{bc}	58.75	71.43
T ₇	Chlorantraniliprole 9.3% + Lambda Cyhalothrin 4.6% ZC	200 ml	1.80 (1.34)	1.00 (0.99) ^b	1.00 (0.99) ^{ab}	0.93 (1.18) ^{ab}	26.19	1.13 (1.06)	0.46 (0.98) ^c	0.40 (0.94) ^{cd}	0.40 (0.94) ^{bc}	50.00	38.09
T ₈	Flubendiamide 480 SC (39.35 w/w)	150 ml	1.60 (1.24)	1.13 (1.05) ^b	0.80 (0.87) ^{abc}	0.73 (1.10) ^{ab}	42.06	1.06 (1.03)	0.86 (1.16) ^{ab}	0.73 (1.11) ^{ab}	0.60 (1.04) ^{ab}	25.00	33.53
T ₉	Control		1.93 (1.38)	1.93 (1.38) ^a	1.33 (1.15) ^a	1.26 (1.32) ^a	-	1.26 (1.12)	0.93 (1.19) ^a	0.80 (1.13) ^a	0.80 (1.13) ^a	-	-
C.D. (P≤0.05)			NS	0.29	0.30	0.23	-	NS	0.18	0.11	0.16	-	-
C.V. (%)			16.67	18.40	21.47	13.11	-	10.63	10.25	7.15	10.20	-	-

*Mean of 5 plants

Figures in parentheses are square root transformed values; Treatments denoted with same alphabets within a column are not significant at 5% level

DAS – Days after Spraying;

Table 3 Accuracy, repeatability and reproducibility of indoxacarb at various fortification levels in immature cobs

Replication	Fortified level (ppb)									
	Accuracy					Repeatability				
	0.01 ppm		0.1ppm		Per cent recovery	0.01 ppm		0.1ppm		Per cent recovery
	Recovered Level (ppb)	Per cent recovery	Recovered Level (ppb)	Per cent recovery		Recovered Level (ppb)	Per cent recovery	Recovered Level (ppb)	Per cent recovery	
R1	0.0103	103.32	0.0974	97.40	0.0098	0.0098	97.93	0.0109	108.70	97.04
R2	0.0099	99.26	0.0964	96.39	0.0096	0.0096	95.56	0.0101	100.86	95.64
R3	0.0103	103.34	0.0964	96.39	0.0101	0.0101	101.13	0.0099	99.05	95.96
R4	0.0097	97.30	0.0961	96.10	0.0091	0.0091	91.30	0.0100	99.68	96.66
R5	0.0099	98.61	0.0967	96.66	0.0098	0.0098	98.21	0.0099	99.16	96.44
Average		100.37		96.59			96.83		101.49	96.35
SD (±)		2.79		0.49			3.67		4.09	0.56
RSD (%)		2.78		0.51			3.79		4.03	0.58

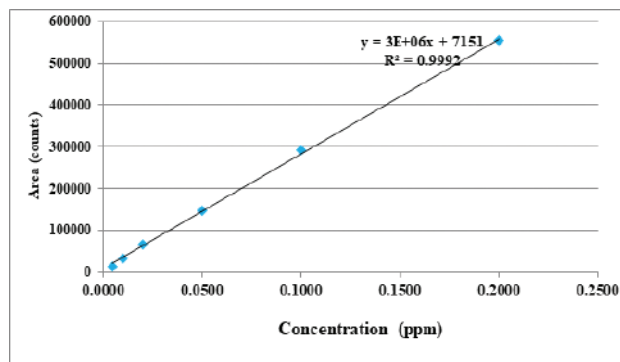


Figure 1.2 Linearity plot of Indoxacarb in immature Cob

Table 4. Dissipation pattern of indoxacarb in immature cob

Days after final spray	Residues of indoxacarb (ppm)				Per cent dissipation
	R1	R2	R3	Mean	
0	0.029	0.028	0.029	0.029	-
1	0.014	0.014	0.017	0.015	48.27
3	<LOQ	<LOQ	<LOQ	<LOQ	100.0
5	<LOQ	<LOQ	<LOQ	<LOQ	100.0
7	<LOQ	<LOQ	<LOQ	<LOQ	100.0
10	<LOQ	<LOQ	<LOQ	<LOQ	100.0
15	<LOQ	<LOQ	<LOQ	<LOQ	100.0
25	<LOQ	<LOQ	<LOQ	<LOQ	100.0
Regression equation	$-0.2863x + 1.7487$				
R ²	1.00				
Half-life (days)	1.05				

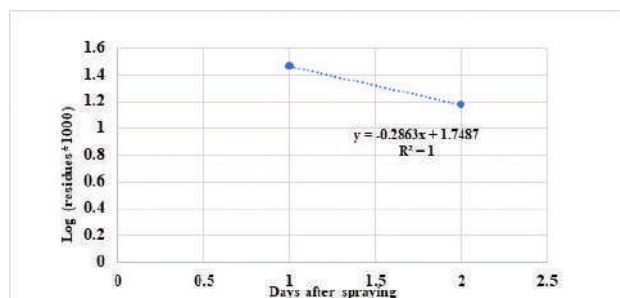


Figure 3. Semi logarithmic graph depicting dissipation of Indoxacarb

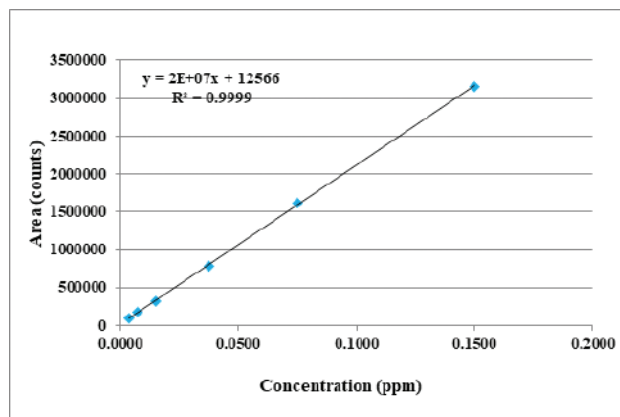


Figure 4. Linearity plot of Spinosyn-J in immature cob

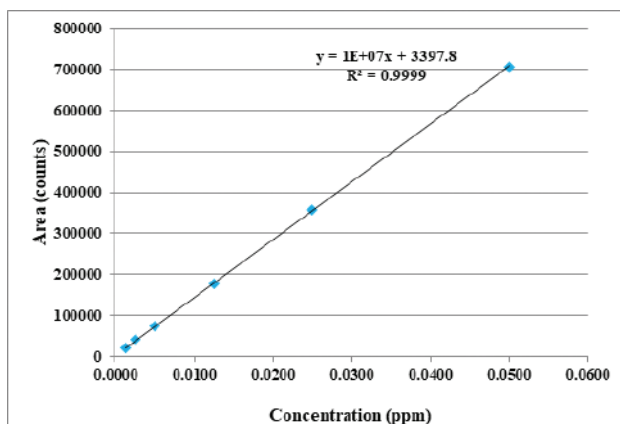


Figure 5. Linearity plot of Spinosyn-L in immature cob

Table 7. Dissipation pattern of spinetoram in immature cob

Days after final spray	Residues of spinetoram (ppm)				Per cent dissipation
	R1	R2	R3	Mean	
0	<LOQ	<LOQ	<LOQ	<LOQ	-
1	<LOQ	<LOQ	<LOQ	<LOQ	100.0
3	<LOQ	<LOQ	<LOQ	<LOQ	100.0
5	<LOQ	<LOQ	<LOQ	<LOQ	100.0
7	<LOQ	<LOQ	<LOQ	<LOQ	100.0
10	<LOQ	<LOQ	<LOQ	<LOQ	100.0
15	<LOQ	<LOQ	<LOQ	<LOQ	100.0
25	<LOQ	<LOQ	<LOQ	<LOQ	100.0
Regression equation				-	
R ²				-	
Half-life (days)				-	

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Determination of economic injury level of mustard aphid (*Lipaphis erysimi*) on Mustard

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Abstract

The Mustard is one of the important crops of rabi season in Kashmir. It provides forage to honey bees in early spring besides oilseed for human consumption. Mustard crop is attacked by various insect pests out of which Mustard aphid (Lipaphis erysimi Kalt.) occupies a predominant position. It thrives well under climatic conditions of Kashmir and both nymphs and adults remain in a damaging stage from late winter to early spring when the crop is at flowering and pod formation stages leading to various deformities in the plant. Management is often restricted to insecticides that are applied indiscriminately without any regard to threshold levels. In this study an experiment was conducted to establish the economic threshold level of mustard aphid in the rabi season of 2023-2024. The crop was subjected to five exposure periods based on the incidence of the pest in the various fields of mustard over the period of years. The findings revealed that economic injury level (EIL) of mustard aphid was worked out to be 72.19 aphids/10 cm apical twig/plant, while as economic threshold level (ETL) was found to be 54.14 aphids/10cm apical twig/plant. There was a strong negative correlation ($r = 0.99$) between aphid density and seed yield of mustard.

Keywords: Mustard, *Lipaphis erysimi*, economic threshold level

Introduction

Mustard *Brassica campestris* (Linn.) Var. Brown Sarson is main *rabi* crop of Kashmir. It is an important oilseed crop grown for edible oil and condiment. The crop is grown over an area of 46,000 hectares in Jammu & Kashmir of which 80% is cultivated in Kashmir. (DOSTAT, 2023). Among the insect pests, Mustard aphid *Lipaphis erysimi* (Kalt.) holds an important place as it attacks the crop from the seedling stage till maturity. The aphid attacks all the aerial parts of the plant like leavers, young shoots, inflorescence and pods (Mishra and Gaurav, 2022). The yield loss due to infestation of mustard aphid accounts for 30-50% due to sucking of sap resulting in stunted growth and development of sooty mould on secreted honey dew that makes harvesting of pods unfeasible (Package of

Practices SKUAST-K). The management is often relied on the calendar application of insecticides without due regard to thresholds. The concept of Economic threshold level (ETL) and Economic injury level (EIL) address the need for initiating the control measures at actual pest density (Pal *et al.*, 2013). The dual concept of Economic threshold level (ETL) and Economic injury level (EIL) intend to harmonize the pest control operation within the ambit of principles of integrated pest management.

Economic injury level (EIL) is defined as the lowest pest population that will impart economic loss to the crop while whereas, Economic threshold level (ETL) is the pest density per unit area of plant at which control measures should be initiated to stop the increasing pest population

from reaching the EIL. In this study, we aim to establish the ETL and EIL of mustard aphid on mustard crop so that judicious pest management decisions are arrived at with respect to the pest.

Material and methods

The present study was conducted in district Ganderbal of Kashmir province with humid western Himalayan subregion. Mustard is a major winter crop in the region. The experiment was laid in the *rabi* season of 2023-2024 using Shalimar Sarson-2 variety. The sowing was done in the 2nd week of October. The crop was raised by following recommended agronomic practices of SKUAST-K. The experiment was carried out in Randomized Block Design having five exposure periods, each replicated thrice. The methodology as proposed by Bakhietia *et al.*, 1979 was used in the present study by adopting different exposure periods. Dimethoate 30 EC @ 1ml/litre of water was sprayed for maintaining five population densities of aphids at 15-day interval. In order to have zero exposure to aphids, four insecticide treatments were given to that treatment.

Treatment having eight-week exposure to aphid incidence was treated as untreated control having no application of insecticide. The number of aphids/10 cm apical twig was observed from the first appearance of aphid infestation in 10 randomly selected plants/replicate. Development of linear regression between aphid density (x) and seed yield at harvest (Y) was computed. Cost T₂ treatment (including insecticide, wages, sprayer and labour charges) and market price of seed were used for obtaining the gain threshold (GT). The ETL was determined as described by Pedigo 1991 while as EIL was determined as per Stone and Pedigo 1972. For sucking insect pests like mites and aphids, where injury and damage are difficult to separate, the regression slope (b), the (b = I × D) represents damage per insect (Pedigo and Rice, 2009). The reduction in seed yield and aphid density at different exposure levels, the economic injury level (EIL) and gain threshold were obtained by the methodology as elaborated by Johnson and Bishop (1987).

The detailed procedure for obtaining Economic Injury Level (EIL) is described below.

1. Regression equation: $Y = (a \pm bx)$

Where x = aphid population (aphid population/10 cm apical twig)

y = seed yield of mustard (kg/ha)

a = intercept on y

b = regression coefficient

2. Gain threshold (kg/ha) : $\frac{\text{Cost of treatment (Rs/ha)}}{\text{Price of mustard seed (Rs/kg)}}$
3. Calculated EIL : $\frac{\text{Gain threshold (kg/ha)}}{\text{Regression coefficient (b)}}$
4. Actual EIL: EIL (calculated) + UAD
UAD = unavoidable aphid density

Results and Discussion

Different exposure period of aphids was given to mustard crop ranging from 0 to 60 days at different standard meteorological weeks. The overall mean aphid infestation/10 cm apical twig ranged from 11.76 to 79.60 in T₁ to T₅ respectively. Least mean aphid infestation being 11.76 was observed when the crop was subjected to 0-week exposure period (T₁).

The increasing aphid incidence was experienced with increasing aphid exposure and highest aphid population density 79.60/10 cm apical twig was observed in T₅ when the crop was eight-week exposure period. Despite having 0-day exposure period, the crop witnessed aphid incidence to the tune of 11.76 and was regarded as the unavoidable aphid density (UAD) (Table 1).

Increased aphid population resulted in decreased seed yield and vice versa. Highest seed yield (12.21 q/ha) was observed in T₁ with 0 exposure to aphids and lowest yield (6.41 q/ha) was recorded in T₅ with eight-week exposure period. For gain threshold aphid exposure period of 2 week with two insecticide sprays was taken into consideration (Table 2).

Market price of mustard seed Rs: 50/kg. market price of 1.0 quintal of seed Rs: 5000/-

The increase in seed yield over control ranged from 1.42 q/ha in T₄ to 5.80 q/ha in T₁ when the crop was under 45 and 0 day aphid exposure respectively. Highest seed yield over control was observed in T₁ and lowest in T₄. The increase in seed yield over control in T₂ and T₃ was recorded as 5.32 and 3.20 q/ha when the crop was under 15 and 30 day aphid exposure respectively (Table 2).

Table 1. Exposure of Mustard crop to aphids under different treatments

Treatments	Aphid exposure to crop (days)	Infestation of aphid on 10 cm apical twig (standard weeks)					Overall Mean
		15	16	17	18	19	
T1	0	15.67 (3.95)*	10.81 (3.28)	9.87 (3.14)	14.13 (3.75)	8.31 (2.88)	11.76
T2	15	16.12 (4.01)	20.22 (4.49)	27.11 (5.20)	35.11 (5.92)	29.66 (5.44)	25.64
T3	30	19.61 (4.42)	24.65 (4.96)	40.11 (6.33)	55.11 (7.42)	50.12 (7.07)	37.92
T4	45	22.31 (4.72)	32.61 (5.71)	65.11 (8.06)	99.62 (9.98)	91.62 (9.57)	62.25
T5	60	30.17 (5.49)	57.85 (7.60)	80.81 (8.98)	130.30 (11.41)	98.91 (9.94)	79.60
	CD	--	4.62	10.26	12.63	7.65	

*Square root transformation

Table 2. Effect of different aphid densities expressed as exposure period on seed yield of Mustard

Treatments	Aphid exposure (days)	Aphid density/10 cm apical twig	Yield of mustard seed (q/ha)	Increase in seed yield over control (q/ha)	Input cost Rs/ha*	Net income Rs/ha
T1	0	11.76	12.21	5.80	2181.82	26,818.18
T2	15	25.64	11.73	5.32	1813.32	24,786.68
T3	30	37.92	9.61	3.20	1462.75	14,537.25
T4	45	62.25	7.83	1.42	1012.21	6087.79
T5	60	79.60	6.41	--	--	--

*Cost of labour, sprayer charges, insecticides

The gain threshold (GT) value was determined to be 36.26 kg/ha. The calculated economic injury level (cEIL) was found to be 60.43 aphids/10 cm apical twig/plant based on linear regression ($Y = a - bx$). The unavoidable aphid density (UAD) of 11.76 was taken into consideration for calculation of actual economic injury level (aEIL) that came out to be 72.19 aphids/10 cm apical twig/plant (sum of UAD and cEIL). The density of pest population at which management measures must be taken to prevent the increasing population to reach the EIL is known as ETL. The ETL is set at 75 per cent of EIL as per the Pedigo 1991. In this study the ETL of 54.14 aphids/10 cm apical twig/plant was derived from the 75 per cent of the actual EIL (aEIL) of 72.19 aphids/10 cm apical twig/plant.

The obtained EIL of 72.19 aphids/10 cm apical twig/plant on mustard crop is specific to temperate conditions of Kashmir similar needs to be confirmed due to lack of similar studies under Kashmir conditions. However, there

are reports from other research workers like Saunakiya and Tiwari (2014) who reported on EIL of 28.33 aphids/10 cm plant. Dubey *et al.* (1998) worked out ETL of Mustard aphid on mustard crop to be 87 aphids/plant. Ahuja and Kalyan (2003), who reported EIL and ETL as 23 and 18 aphids respectively/10 cm of inflorescence of central shoot of a plant. Singh and Sachan (1997), observed the EIL value of 48 aphids/10 cm terminal shoot/plant on Mustard crop in Tarai region of India.

The present study EIL of Mustard aphid on Mustard crop reveal that management measures should be initiated as soon as the aphid population reaches 54.14 aphids/10 cm apical twig/plant to prevent the pest population breaching the EIL of 72.19 aphids/10 cm apical twig/plant so that economic loss is not experienced by the farmers. Pertinent to mention that economic threshold levels are determined by the prevailing cost of plant protection, market price of crop and agro climatic conditions.

Table 3. Calculation of economic threshold level based of mustard aphid on mustard crop in Kashmir

Exposure period (weeks)	Mean aphid density	Seed yield (q/ha)	Intercept (a)	Gain threshold (kg/ha)	Regression coefficient (b)	Correlation coefficient (r)	Unavoidable aphid density (UAD)	Y=a+bx	EIL	E T L
									Calculated	Actual
2	25.64	11.73	14.08	36.26	-0.06	-0.98	11.76	11.73=14.08+ (-0.06x)	60.43	72.19 54.14

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Occurrence of banana bunchy top virus in tissue-cultured and conventional banana cultivars in the Siang region of Arunachal Pradesh, India

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Banana bunchy top virus (BBTV; genus *Babuvirus*, family *Nanoviridae*) is one of the most destructive viral pathogens of banana (*Musa* spp.), causing severe stunting, bunching, and yield losses that may reach up to 100% in highly susceptible cultivars (Nelson and Thomas, 2020). The virus is transmitted persistently by the banana aphid *Pentalonia nigronervosa* Coquerel and through infected planting material, including tissue-cultured plants when proper virus indexing is not followed (Jebakumar *et al.*, 2018). Although banana is an important fruit crop in Arunachal Pradesh, systematic information on BBTV occurrence in the Siang region is limited. The present study documents the occurrence, incidence, and molecular confirmation of BBTV in tissue-cultured and conventional banana cultivars in this region.

A systematic orchard survey was conducted during August-December 2025 in the Siang region of Arunachal Pradesh, covering Likabali (~240 m AMSL), Aalo (~300 m AMSL), and Basar (~650 m AMSL). Banana plants were examined for characteristic BBTV symptoms, including marginal chlorosis, dark green discontinuous streaks (“Morse code”) on the midrib and veins, J-shaped hooks, narrow upright leaves, bunching, and severe stunting. Disease incidence was calculated as the percentage of symptomatic plants relative to the total number of plants observed at each location.

Both tissue-cultured banana cv. Grand Nain and conventionally propagated cultivars (Dwarf Cavendish, Giant Cavendish, Monthan [Kas Kol], and Poovan) were surveyed. Wild *Musa* species (*M. rosacea* Jacq. and *M.*

velutina Wendl.) occurring naturally in the region were also examined. Young leaf samples from symptomatic and asymptomatic plants were collected, and total genomic DNA was extracted using the CTAB method. PCR amplification targeting the BBTV DNA-S (coat protein) component was carried out following Manickam *et al.* (2002). Amplified products were resolved on 1.2% agarose gel and visualized under UV illumination. PCR amplification using BBTV DNA-R-specific primers was performed to further confirm BBTV infection. PCR amplification yielded the expected ~1.0 kb amplicon in all symptomatic samples, whereas asymptomatic plants and healthy controls showed no amplification, thereby confirming the presence of Banana bunchy top virus in the surveyed banana cultivars.

Characteristic BBTV symptoms were observed in several banana cultivars under natural orchard conditions. In tissue-cultured banana cv. Grand Nain (Fig. 1), disease incidence ranged from 13.0% to 22.0% across locations, with a pooled incidence of 17.66% (Table 1). Infected plants were predominantly 4-6 months old, indicating early establishment of infection. Among conventional cultivars, Dwarf Cavendish showed very high susceptibility (Fig. 2), with disease incidence reaching 95.0% at Likabali, followed by Basar (87.5%) and Aalo (62.5%). Giant Cavendish and Monthan (Kas Kol) exhibited moderate infection levels (37.5-45.0%), whereas Poovan remained completely free from symptoms. Wild *Musa* species (*M. rosacea* and *M. velutina*) also showed no symptoms and tested negative by PCR (Table 2).

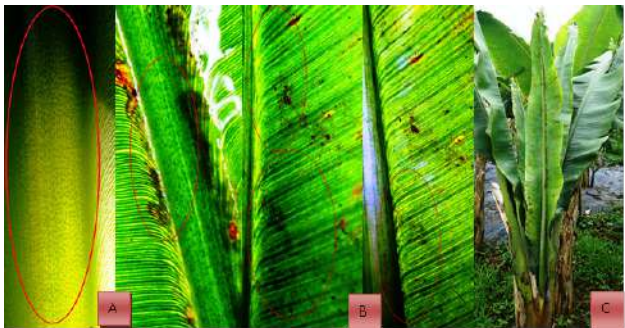


Figure 1. Symptoms of BBTV on tissue-cultured banana cv. Grand Nain: (A) Morse code streaking on leaf midrib; (B) J-shaped hooks along the midrib; (C) bunching symptom

Table 1. Incidence of BBTV in tissue-cultured banana (cv. Grand Nain) under natural orchard conditions in the Siang region, Arunachal Pradesh (Aug-Dec 2025)

Location	Total plants surveyed	Symptomatic plants	Disease incidence (%)	Age of infected plants (months)	Source of planting material	PCR confirmation
Likabali	100	22	22.0	4-6	Commercial nurseries, Assam (India)	Positive
Aalo	100	18	18.0	4-6		Positive
Basar	100	13	13.0	4-6		Positive
Total	300	53	17.66			

Table 2. Reaction of banana cultivars and wild *Musa* species against BBTV under natural orchard conditions in the Siang region, Arunachal Pradesh

Variety / Cultivar	Location	Total plants observed	Symptomatic plants	Disease incidence (%)	Age of infected plants (months)	Source of planting material
Dwarf Cavendish	Likabali	40	38	95.0	5-6	Commercial nurseries, Assam (India)
	Aalo	40	25	62.5	5-6	
	Basar	40	35	87.5	5-6	
Giant Cavendish	Basar	40	15	37.5	3-5	Wild sources
Monthan (Kas Kol)	Basar	40	18	45.0	4-6	
Poovan	Basar	40	0	0.0	5-6	
<i>Musa rosacea</i> Jacq.	Likabali	40	0	0.0	6-7	Wild sources
<i>Musa velutina</i> Wendl.		40	0	0.0	6-7	

PCR amplification of BBTV DNA-S and DNA-R genomic components consistently produced the expected amplicons in symptomatic samples, while asymptomatic plants remained negative (Fig. 2), confirming BBTV infection. The detection of BBTV in tissue-cultured Grand Nain highlights the risk associated with inadequate virus indexing of planting material, whereas the extremely high incidence in Dwarf Cavendish reflects widespread exposure and efficient aphid-mediated transmission. The absence of infection in Poovan and wild *Musa* species suggests possible

resistance or escape under natural orchard conditions, as reported earlier for certain banana genotypes (Nelson and Thomas, 2020). This study confirms the occurrence of Banana bunchy top virus in both tissue-cultured and conventional banana cultivars in the Siang region of Arunachal Pradesh. The findings emphasize the urgent need for the use of certified virus-free planting material, routine PCR-based surveillance, effective aphid management, and prompt removal of infected plants to prevent further spread and safeguard banana production in the region.

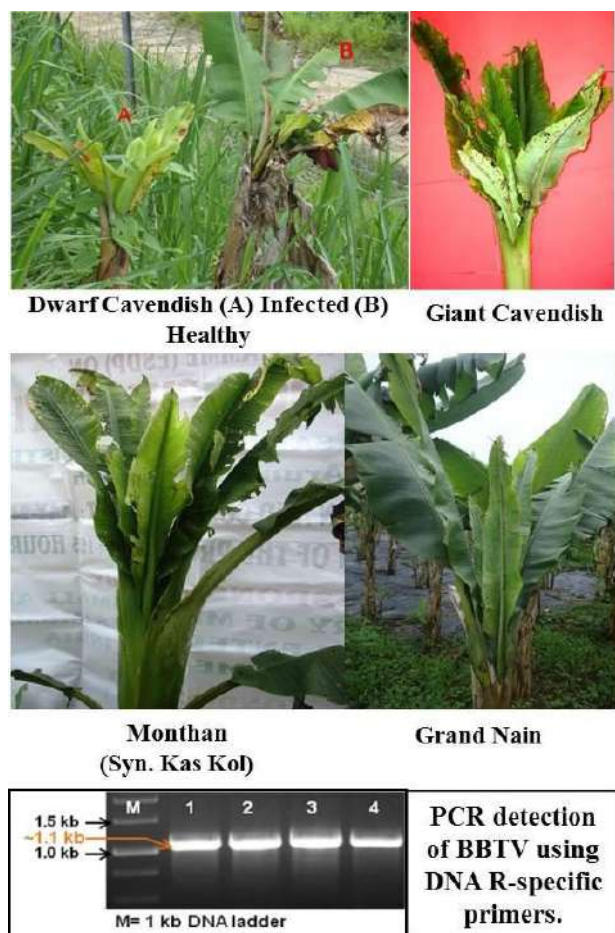


Figure 2. Severe BBTV symptoms on Dwarf Cavendish, Giant Cavendish, Monthan and Grand Nain and PCR-based detection of BBTV using DNA-S and DNA-R-specific primers. Lanes 1-4: symptomatic samples; M: 1 kb DNA ladder; expected amplicons (~1.1 kb for DNA-S and ~1.0 kb for DNA-R)

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